Biomolecular dynamics through latent ensemble decomposition with OriDynamic

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Abstract

Biomolecular function depends on conformational variability across structural states and fluctuations within them. Learning this variability requires a representation of how an ensemble is organized, beyond the plausibility of its individual conformations. Here, we introduce OriDynamic, a generative framework for biomolecular ensembles through latent ensemble decomposition. Our formulation organizes conformational variability into representative structural states and conditional distributions of atomic configurations around those states. An autoencoder learns a latent structural representation, a flow module generates representative conformations across basins, and a refine module performs conditional all-atom sampling to explore variation within each basin while improving local geometry. Complementary ensemble-level objectives constrain mean atomic positions, fluctuation variances and joint coordinate covariance, connecting the organization of generation to the collective statistics of its output. Across four public benchmarks covering protein monomers, protein–ligand complexes and protein–protein assemblies, OriDynamic achieves leading overall performance in reproducing reference ensembles. Three application studies further demonstrate the recovery of functionally relevant alternative states, partner-dependent apo and holo conformations, and configurations associated with ligand dissociation. By explicitly coordinating the exploration of alternative states with the sampling of their internal variability, OriDynamic provides a basis for incorporating conformational ensembles into AI-driven drug discovery. Future integration with large language models could further connect scientific reasoning to explicit representations of molecular variability and its functional implications.

1 Introduction

Biomolecular function emerges from the conformations molecules access, their relative populations and the atomic fluctuations within each state. Alternative conformations enable catalysis and molecular recognition [1, 2], while collective fluctuations contribute to conformational entropy and allosteric regulation [3–5]. Interactions with molecular partners can select pre-existing states or induce structural adaptation, reshaping both the configurations available to a molecule and the fluctuations around them [6]. For AI-driven drug discovery, these properties determine the structural possibilities available for molecular intervention: transient states can expose binding pockets, and changes in local geometry and coordinated motion can alter how a target accommodates a ligand or another protein [7, 8]. Modeling these possibilities requires a description of conformational ensembles that captures both alternative structural states and the variability associated with them.

Static biomolecular structure prediction has advanced rapidly in accuracy, molecular coverage and accessibility. Building on the breakthroughs established by AlphaFold 2 and AlphaFold 3 [9, 10], several complementary developments have expanded its capabilities. Protein-language-model-based approaches, exemplified by ESMFold [11] and ESMFold2 [12], use representations learned from sequence data to support accurate structure prediction, with the latter extending this approach to all-atom biomolecular complexes. An expanding ecosystem of AF3-like cofolding models—including Protenix-v1 and Protenix-v2 [13], and Boltz-1 and Boltz-2 [14]—has further advanced complex prediction and introduced capabilities such as template conditioning, contact constraints

and, in Boltz-2, binding-affinity prediction [15]. Open releases of Protenix-v1 and the Boltz models have also broadened access to these capabilities. More recently, IsoDDE, OpenDDE and the AnewFold component of AnewDDE have reported substantial gains in antibody–antigen structure prediction, extending the reach of structural modeling to challenging molecular interfaces [16–18].

These advances provide an increasingly powerful structural foundation for molecular discovery. They also sharpen the distinction between predicting accurate structures and learning a conformational ensemble. Modifying evolutionary inputs or sampling different random seeds can produce alternative conformations for the same molecular system [10, 19]. Their individual plausibility, however, does not establish whether the resulting collection reproduces the target conformational distribution. Ensemble modeling additionally requires these conformations to collectively capture accessible states, fluctuations within those states and the joint statistics of atomic displacements. The central modeling question therefore extends from which structures can be generated to how conformational variability is represented, organized and learned.

Molecular dynamics provides an atomistic account of conformational fluctuations and transitions, although sampling separated conformational basins remains computationally demanding [20]. Generative approaches have developed complementary strategies for learning equilibrium ensembles and temporal evolution. AlphaFlow adapts structure predictors to ensemble generation [21]. BioEmu and its successor BioEmu2 advance ensemble emulation from proteins toward broader biomolecular systems: BioEmu incorporates thermodynamic calibration, while BioEmu2 extends molecular coverage and supports both ensemble sampling and time-lagged propagation [22, 23]. AnewSampling targets all-atom equilibrium distributions of protein–ligand interactions [24], whereas MDGen and BioKinema model temporal relationships through trajectory generation [25, 26]. These developments broaden the molecular scope and generative targets of biomolecular AI. A complementary modeling question concerns how conformational variability should be organized within a generator, so that exploration across distinct states and sampling within them can be learned together.

Our starting point is that the representation of an ensemble should make this organization explicit. Distinct conformational basins contain alternative structural arrangements, and each basin encompasses a distribution of atomic configurations around those arrangements. We therefore represent conformational variability through representative structural states and the distributions of configurations conditioned on them. In this formulation, a representative conformation serves both as an accessible structural state and as a condition for generating further variability. This separates the task of discovering alternative states from that of resolving fluctuations within them, while retaining both as parts of a common ensemble. Recovering representative states alone is insufficient: ensembles with similar state coverage can still differ in the magnitude, directionality and coordination of their atomic fluctuations. The generative organization must therefore be accompanied by constraints on the collective statistics of its output.

Guided by this view, we introduce OriDynamic, a framework for biomolecular ensemble generation through latent ensemble decomposition. The framework implements two complementary decompositions: it organizes generation into variation across basins and fluctuations within them, and it organizes statistical supervision into objectives for mean positions, variances and joint covariance. Built on an AlphaFold 3–like cofolding architecture, OriDynamic comprises three modules: an autoencoder, a flow module and a refine module. The autoencoder first learns a latent representation through structural encoding and reconstruction. The flow module then learns to generate representative conformations across basins in this latent space. Conditioned on the decoded representatives, the refine module performs stochastic all-atom sampling to explore variability within each basin while improving local geometry. Together, these modules translate the chosen ensemble representation into a conditional generative process.

Ensemble-level objectives connect this generative decomposition to the statistics of reference ensembles. Mean and variance objectives constrain average atomic positions and fluctuation amplitudes, while a covariance objective constrains the full, trace-normalized joint coordinate covariance of selected representative atoms. This joint covariance captures both the directional variability of individual atoms and correlated displacements between different atoms. Its normalization emphasizes the relative structure of joint fluctuations, complementing the amplitude information supplied by the variance objective. These constraints are applied to the final generated ensemble, making collective variability an explicit target of learning. A common architecture supports protein monomers, protein–ligand complexes and protein–protein assemblies, using a jointly trained model supervised by ATLAS, MISATO and mdCATH conformational ensembles; ligand-dissociation evaluation uses the DD-13M adaptation described in Methods. OriDynamic thus generates time-independent ensembles conditioned on molecular context.

We evaluate OriDynamic on four public benchmarks spanning protein monomers (ATLAS [27] and mdCATH [28]), protein–ligand complexes (MISATO [29]) and protein–protein assemblies (the DynaPPI benchmark from DynaRepo [30]). Across these benchmarks, OriDynamic achieves leading overall performance in reproducing reference ensembles, including their conformational flexibility and interaction geometry. Three application studies further demonstrate the biological relevance of the generated ensembles: OriDynamic recovers functionally relevant alternative states, captures partner-dependent apo and holo conformations, and generates configurations associated with ligand dissociation. These results support the applicability of latent ensemble decomposition across diverse molecular systems and connect alternative structural states to binding-associated adaptation and changes in intermolecular contacts, providing structural hypotheses for investigating molecular recognition and the conformational basis of biomolecular function.

Looking ahead, integrating OriDynamic with large language models (LLMs) could connect scientific reasoning to an explorable representation of molecular variability. Generated ensembles and their quantitative descriptions could help LLMs reason about alternative structural states, compare interaction geometries and examine coordinated atomic fluctuations. In turn, LLMs could use scientific knowledge to guide conformational queries, formulate mechanistic hypotheses and propose experiments that distinguish competing explanations. Coupling this interaction with physical modeling and experimental feedback could help ground reasoning about the microscopic basis of molecular function in explicit, testable structural predictions. Such integration offers a path toward AI systems that connect scientific knowledge with molecular exploration, advancing drug discovery and the investigation of biological mechanisms.

2 Results

2.1 Overview

OriDynamic organizes ensemble generation into the sampling of representative conformations across basins and the conditional exploration of variability within them (Fig. 1f). This formulation represents an ensemble through alternative structural states and the distributions of atomic configurations conditioned on those states. It is implemented through three modules—an autoencoder, a flow module and a refine module—integrated with an AF3-like cofolding backbone (Fig. 1c). A common architecture supports protein monomers, protein–ligand complexes and protein–protein assemblies, using a jointly trained model (Fig. 1a).

The trunk encodes molecular context into single and pair representations. During training, the autoencoder maps trajectory conformations to latent representations relative to a structural anchor and learns to reconstruct their coordinates. The anchor is supplied as an input structure or predicted by OriDynamic's folding backbone. Conditioned on molecular context, the flow module learns to generate latent representations from noise, which are decoded into representative conformations across basins. The refine module then uses coordinate diffusion to perform stochastic all-atom sampling conditioned on these decoded representatives, exploring configurations within each basin while improving local geometry. Repeated sampling of representatives and their conditional fluctuations produces an ensemble for the supplied molecular context; changing interaction partners enables comparison of ensembles under different contexts.

Training proceeds in three stages (Fig. 1e). First, the autoencoder learns the structural latent representation through encoding and reconstruction. Second, the autoencoder is frozen while the flow module learns conformational variation in the fixed latent space. Third, the flow and refine modules are jointly fine-tuned together with low-rank adaptation (LoRA) parameters in the trunk, while the autoencoder remains frozen. This schedule establishes the representation before learning the generation of representative states and their conditional atomic variability.

A complementary decomposition organizes ensemble-level supervision into objectives for mean coordinates, fluctuation variances and joint coordinate covariance (Fig. 1c). For selected representative atoms, the covariance objective constrains the full joint covariance of their flattened three-dimensional displacement vectors, normalized by its trace. It therefore captures both directional variability within individual atoms and cross-covariance between different atoms. The variance objective supplies complementary constraints on fluctuation amplitudes. These losses are evaluated on the final combined ensemble, linking the two generative stages to a shared set of statistical targets. Trajectory frames provide structural examples and ensemble statistics for a time-independent prediction task.

Joint training uses 7,260 targets from ATLAS, MISATO and mdCATH: 3,196 protein monomers or domains and 4,064 protein–ligand complexes. The model is additionally adapted on 144 DD-13M systems for ligand-dissociation evaluation. Protein–protein ensemble performance is assessed on 36 DynaPPI test complexes (Fig. 1d; Methods, Data). The following sections evaluate ensemble fidelity on ATLAS, mdCATH, MISATO and DynaPPI

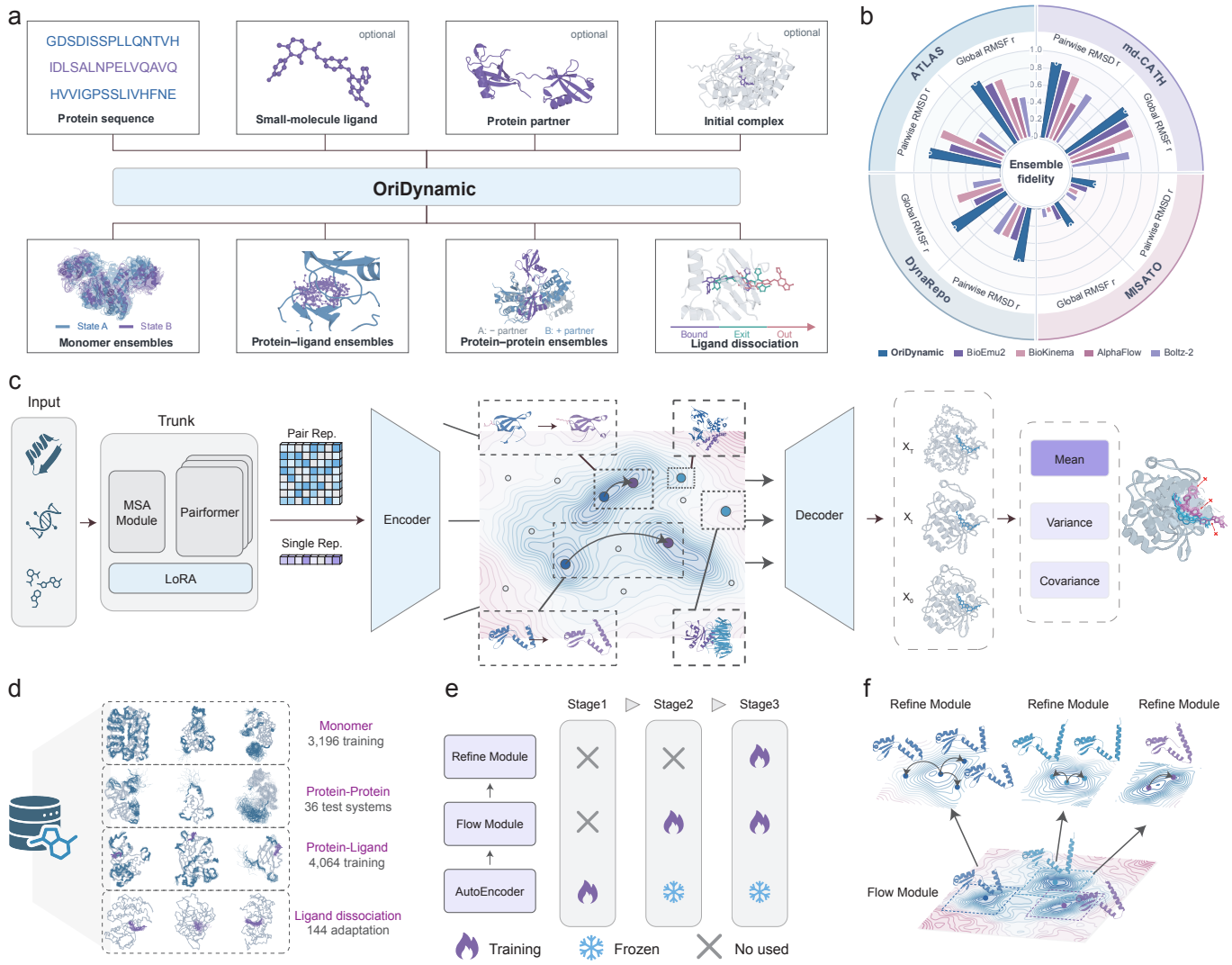


Figure 1. OriDynamic and latent ensemble decomposition. a, Molecular inputs and supported ensemble-generation tasks. **b,** Ensemble fidelity on ATLAS, mdCATH, MISATO and DynaPPI, summarized by RMSF and pairwise-RMSD correlations. **c,** A molecular trunk conditions an autoencoder, a flow module and an all-atom refine module. Ensemble losses constrain mean coordinates, fluctuation amplitudes and trace-normalized joint covariance. **d,** Data used for joint training (3,196 monomers/domains and 4,064 protein–ligand complexes), ligand-dissociation adaptation (144 systems) and protein–protein evaluation (36 complexes). **e,** Three-stage training; flame, snowflake and cross symbols denote trainable, frozen and unused modules. **f,** Latent sampling generates representative conformations across basins, followed by conditional all-atom sampling within each basin.

(Fig. 1b), and examine functionally relevant alternative states, partner-dependent apo and holo conformations, and ligand dissociation-related configurations (Fig. 1a). Architecture, sampling, data preparation and training details are provided in Methods.

2.2 OriDynamic captures conformational variability in protein monomers

We first examined monomer ensembles on ATLAS and mdCATH, comparing ensemble breadth, spatial fluctuation patterns and projected conformational distributions. The evaluation uses 82 ATLAS targets and 100 mdCATH targets shared by all ten methods, with 250 predicted structures per target. A common Cα atom correspondence, alignment procedure and scoring implementation are applied throughout. The three headline metrics distinguish cross-target agreement in mean pairwise RMSD, correlation of pooled RMSF values and the median within-target RMSF correlation (Fig. 2a,b). OriDynamic is represented by a single prediction version, while BioEmu2 equilibrium ensembles and MD-initialized BioKinema trajectories retain their respective generation protocols.

ATLAS. OriDynamic achieved pairwise-RMSD, global-RMSF and median per-target RMSF correlations of 0.834, 0.833 and 0.892, respectively, the highest point estimates among the displayed methods (Fig. 2a). The strongest comparator for pairwise RMSD was BioKinema (0.769), whereas BioEmu2 had the next-highest global

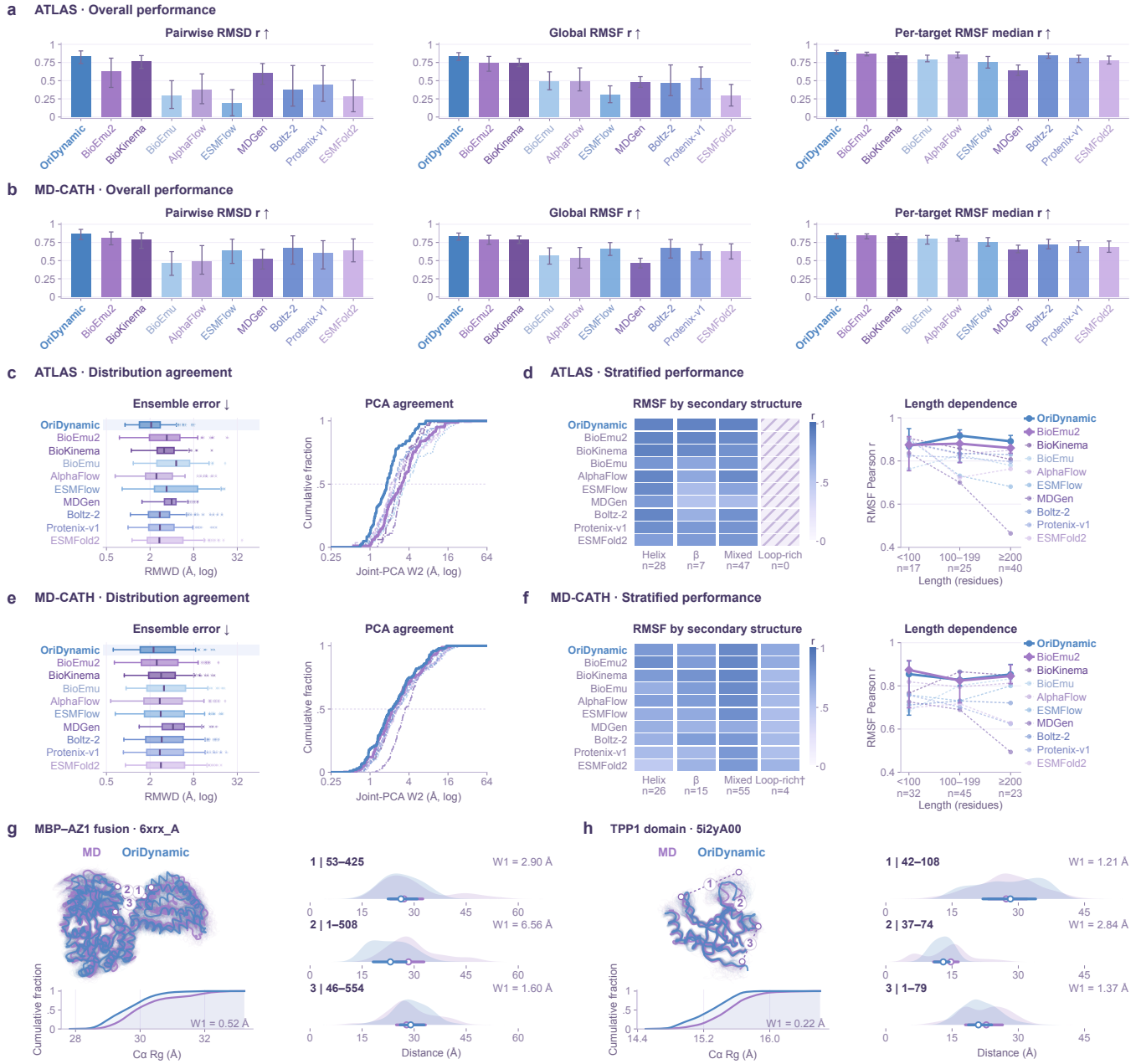


Figure 2. Monomer ensemble fidelity on ATLAS and mdCATH. Ten methods use the same 82 ATLAS and 100 mdCATH targets, with 250 predictions per target and 301/400 MD reference frames. **a,b,** Pairwise-RMSD, pooled-RMSF and median per-target RMSF correlations. **c,e,** RMWD distributions and joint-PCA W_2 cumulative distributions; lower values indicate closer agreement. Boxes show medians and interquartile ranges, with 1.5-IQR whiskers. **d,f,** RMSF correlations by MD-derived secondary structure and sequence length; hatching denotes empty groups and † indicates $n < 5$. **g,h,** Selected 6xrx_A and 5i2yA00 ensembles, residue-pair distance distributions and radii of gyration (MD, purple; OriDynamic, blue). Summary error bars are 95% target-bootstrap intervals. Case statistics use all stored conformations; structure traces are subsampled for display. Evaluation details are in Appendix C.2.

and per-target RMSF correlations (0.747 and 0.871). Ensemble distances provided a complementary assessment of absolute distributional agreement (Fig. 2c). OriDynamic had median RMWD and Joint-PCA W_2 values of 2.073 and 1.814 Å, compared with 2.447 and 2.119 Å for AlphaFlow, the closest comparator under these two summaries. Thus, its association with MD flexibility was accompanied by smaller median errors in coordinate statistics and projected ensemble distributions.

mdCATH. Across the 100 targets, OriDynamic achieved correlations of 0.870, 0.836 and 0.847 for the same three metrics (Fig. 2b). BioEmu2 reached 0.816, 0.789 and 0.841, and BioKinema reached 0.784, 0.790 and 0.834. OriDynamic also had the lowest median RMWD and Joint-PCA W_2 values, 2.233 and 2.172 Å, followed by BioEmu2 at 2.474 and 2.219 Å (Fig. 2e). The three correlations and both ensemble-distance metrics consistently placed OriDynamic among the leading methods on mdCATH.

Dependence on structural composition and length. Reference-derived secondary-structure groups reveal variation that is obscured by dataset-wide summaries (Fig. 2d,f). On ATLAS, OriDynamic's median RMSF correlations were approximately 0.892–0.893 across the three populated composition groups. mdCATH showed stronger agreement for mixed helix/strand targets (0.863) than for helix-dominant (0.780) or strand-dominant targets (0.752); other methods led in several individual groups. The Loop-rich category contains no ATLAS targets and four mdCATH targets. Across the three length bins, OriDynamic's medians ranged from 0.869 to 0.917 on ATLAS and from 0.828 to 0.855 on mdCATH, without a monotonic decrease with length. These analyses support broad agreement while retaining target-class and method-specific differences.

Illustrative conformational distributions. The selected 6xrx_A and 5i2yA00 examples compare structural ensembles with distributions of radius of gyration and three inter-residue distances (Fig. 2g,h). Radius-of-gyration W_1 distances were 0.52 and 0.22 Å, respectively. For the selected inter-residue distances, the 1–508 pair in 6xrx_A had $W_1 = 6.56$ Å, and the 37–74 pair in 5i2yA00 had $W_1 = 2.84$ Å. Together, the benchmark and case analyses show agreement with MD in conformational breadth, residue-level flexibility and projected ensemble distributions.

2.3 OriDynamic captures ligand flexibility and protein–ligand interactions

To assess whether OriDynamic reproduces the structural variability of protein–ligand complexes, we evaluated 255 MISATO test targets against molecular dynamics (MD) reference ensembles. OriDynamic, BioEmu2, BioKinema, Boltz-2, Protenix-v1 and ESMFold2 were compared using 100 structures per method and target and 100 MD frames, with matched atoms and a common alignment protocol (Fig. 3).

OriDynamic attained the highest point estimates for all three ligand-only correlations (Fig. 3a–c). The across-target correlation of mean pairwise root-mean-square deviation (RMSD) was 0.275 (95% CI, 0.169–0.378), compared with 0.205 for BioEmu2 and 0.202 for ESMFold2. The median within-target atom-wise root-mean-square fluctuation (RMSF) correlation was 0.695 (95% CI, 0.663–0.771), versus 0.573 for BioEmu2. Global RMSF correlation, computed after pooling ligand atoms across targets, was 0.303, compared with 0.198 for ESMFold2, the closest comparator. These three metrics show improved agreement with MD in both within-ligand fluctuation patterns and variation across complexes.

Across the eight interaction classes shown in Fig. 3d, OriDynamic had the lowest median Wasserstein distance in six: hydrogen-bond acceptors and donors, anionic interactions, π–cation contacts, π-stacking and hydrophobic interactions. It also had lower median distances than BioEmu2 and BioKinema in all eight classes. Boltz-2 performed best for cationic and cation–π interactions, indicating that the relative advantage depended on interaction type.

OriDynamic also more closely reproduced the magnitude of ligand variability (Fig. 3e). Ratios of the across-target predicted and MD medians were 1.178 for pairwise spread and 1.256 for RMSF amplitude. BioEmu2 and BioKinema overestimated both measures, whereas Boltz-2, Protenix-v1 and ESMFold2 underestimated them. Within 10 Å binding pockets, OriDynamic achieved a median atom-wise RMSF correlation of 0.705, compared with 0.563 for BioEmu2. Across-target pocket pairwise RMSD and mean RMSF correlations were 0.437 and 0.540, respectively (Fig. 3e,f), supporting improved recovery of local fluctuation patterns and overall pocket flexibility.

The 2XZG and 5L9L examples relate these ensemble metrics to individual ligand-binding environments (Fig. 3h,i). Pocket RMSF correlations were 0.598 and 0.768, with mean amplitudes approximately 20% above and 8% below MD, respectively. In a common MD-fitted principal component analysis (PCA) basis, predicted-to-MD spread

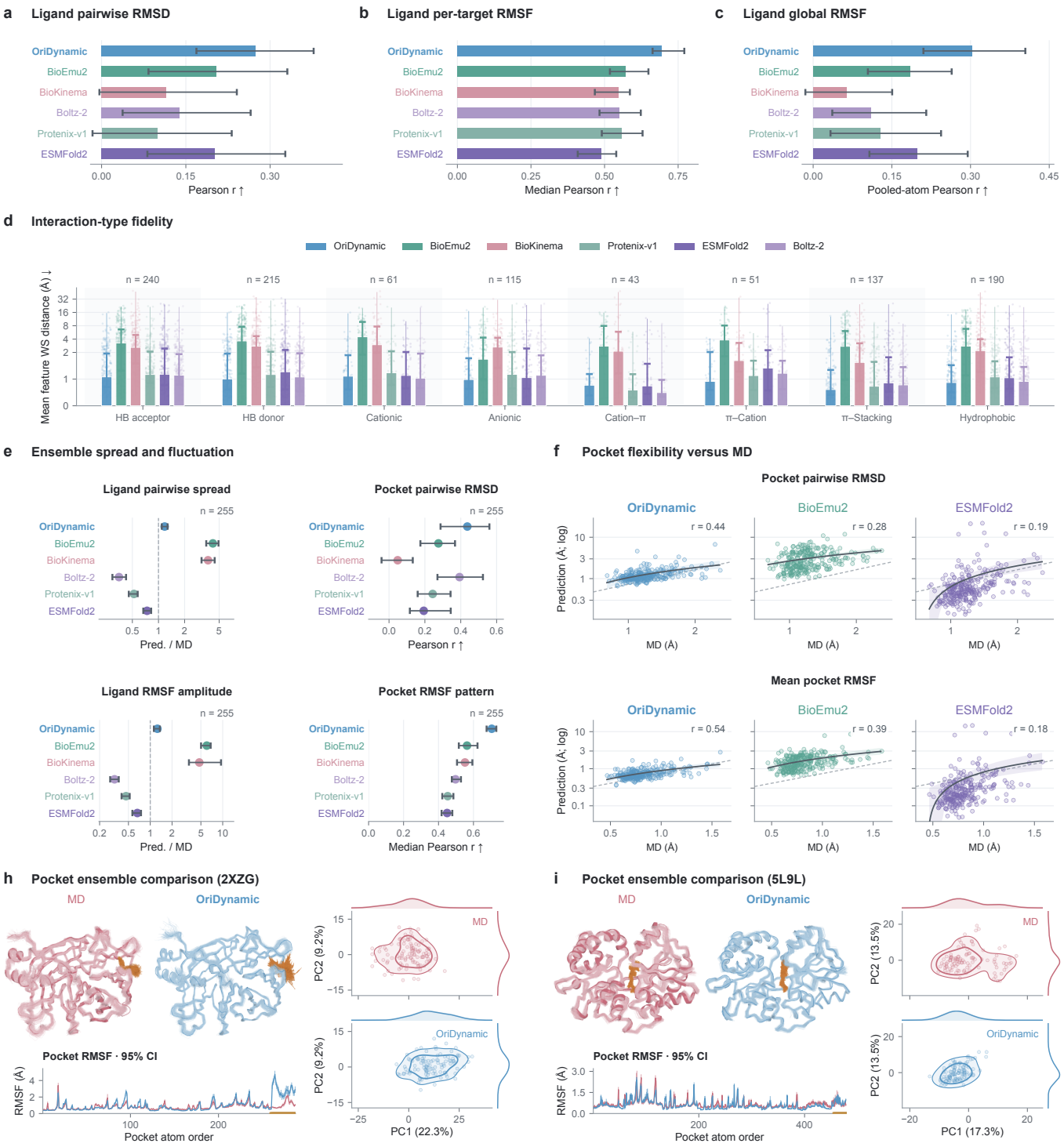


Figure 3. Protein–ligand ensemble fidelity on MISATO. Six methods are compared on 255 complexes using 100 predictions and 100 MD frames per target. **a–c,** Ligand pairwise-RMSD, median per-target RMSF and pooled-RMSF correlations. **d,** Interaction-distance Wasserstein errors: bars show target medians, thick intervals interquartile ranges, thin whiskers observed ranges and dots individual targets. **e,** Ligand fluctuation amplitudes and pocket correlations. **f,** Predicted versus MD pocket flexibility, with regression lines and 95% mean-fit bands. **h,i,** Selected 2XZG and 5L9L ensembles, pocket RMSF profiles and MD-fitted PCA densities (MD, pink; prediction, blue; ligand, orange). Summary error bars show 95% paired target-bootstrap intervals; profile bands describe conditional sampling uncertainty. Cases were selected for favorable agreement. Atom selections and statistical definitions are given in Appendix C.3.

ratios were 1.057 and 0.604, showing near-matched projected spread for 2XZG but a narrower distribution for 5L9L. The first two PCs explained 31.5% and 30.8% of MD variance. Across the benchmark, the ligand and pocket analyses support improved recovery of fluctuation patterns and interaction-distance distributions.

2.4 OriDynamic captures ensemble heterogeneity at protein–protein interfaces

We next tested whether ensemble fidelity extended to protein–protein complexes, where fluctuations alter contacts and solvent exposure at the binding interface. We evaluated OriDynamic against BioEmu2, BioKinema, Boltz-2, Protenix-v1 and ESMFold2 on the same 36 DynaPPI test complexes, using 100 predicted conformations and 100 reference MD frames per target. Analyses of whole complexes and fixed 10 Å interfaces assessed agreement in ensemble breadth and local fluctuation patterns (Fig. 4a,b).

Across whole complexes, OriDynamic had the highest point estimates for all three correlation metrics. The cross-target correlation of mean pairwise RMSD was 0.620, compared with 0.425 for Boltz-2, the strongest comparator on this metric. The median per-target RMSF correlation was 0.786, compared with 0.774 for BioEmu2, and the correlation of concatenated RMSF profiles was 0.718 (95% confidence interval [CI], 0.668–0.772), compared with 0.541 for BioKinema. At the interface, OriDynamic also had the highest global RMSF correlation (0.603 versus 0.505 for BioKinema). BioEmu2 performed better on interface pairwise RMSD correlation (0.647 versus 0.536) and median per-target RMSF correlation (0.772 versus 0.750). These comparisons distinguish whole-complex variability from interface-specific fluctuation patterns.

Interface-specific metrics further resolved agreement in fluctuation amplitude, solvent exposure and conformational coverage (Fig. 4c,d). OriDynamic had the lowest mean interface RMSF absolute error, 0.801 Å (95% CI, 0.570–1.101 Å), compared with 1.221 Å for BioKinema. Its median exposed-residue Jaccard score was 0.574, compared with 0.540 for BioEmu2, and its median joint-PCA Wasserstein-2 distance was 2.337 Å, compared with 2.506 Å for BioEmu2. These were the strongest comparator point estimates for the respective metrics. For transient-contact recovery, OriDynamic's median Jaccard score was 0.278, compared with 0.345 for BioEmu2 and lower values for the other four baselines.

Eight distinct cases, selected post hoc to illustrate their assigned analyses, relate these benchmark statistics to specific interface properties. In A00C7 and A00CE, solvent-exposure mutual-information maps showed Spearman correlations of 0.625 and 0.628 between prediction and MD (Fig. 4c). Prediction and MD shared 19 dynamically exposed residues in A00C7, yielding a Jaccard score of 0.792. A00CE showed complete overlap across its three dynamically exposed residues. The interface RMSF profiles of A00CC and A00C5 correlated with MD at 0.793 and 0.899, respectively, while retaining differences in the amplitudes of individual fluctuation peaks.

Contact maps for A00BU and A00CI distinguished contacts newly populated relative to the input structure from initially present contacts that were incompletely retained (Fig. 4d). Transient-contact Jaccard scores were 0.621 and 0.400, and weak-contact scores were 0.591 and 0.568, respectively. Complementary projections of A00BS and A00CU into a shared MD-fitted PCA basis showed overlapping regions of conformational space, with residual differences in population density and spread. These case projections describe agreement along the displayed MD modes; the benchmark-level Wasserstein-2 metric uses a joint PCA basis.

The DynaPPI results support recovery of multiple aspects of protein–protein ensemble variability, including fluctuation patterns, solvent exposure and conformational breadth. Across the 36 complexes, OriDynamic combined the highest whole-complex correlation estimates with the lowest mean interface RMSF error and median joint-PCA distance.

3 Applications

We examined function-related conformational changes, partner-dependent structural responses, and ligand-dissociation ensembles using the source-specific reference and sampling protocols described in Appendix C.

3.1 OriDynamic samples function-related alternative conformations

We first compared OriDynamic with Boltz-2, Protenix, AlphaFlow, BioEmu, BioEmu2 and ESMFold2 on 91 ConforMix targets, using 100 evaluated structures per target and method. Dual-end success measures coverage of both experimental reference states under the TM-score or RMSD criterion (Fig. 5a,b). OriDynamic used the no_anchor/Joint11500 model, and BioEmu2 used equilibrium sampling.

OriDynamic achieved the highest observed overall success rates: 0.648 (59/91) for TM-score and 0.407 (37/91) for RMSD. The corresponding rates were 0.604 and 0.374 for BioEmu, 0.462 and 0.187 for BioEmu2, and 0.407 and 0.341 for ESMFold2. OriDynamic covered both states of 9/13 membrane transporters by TM-score and

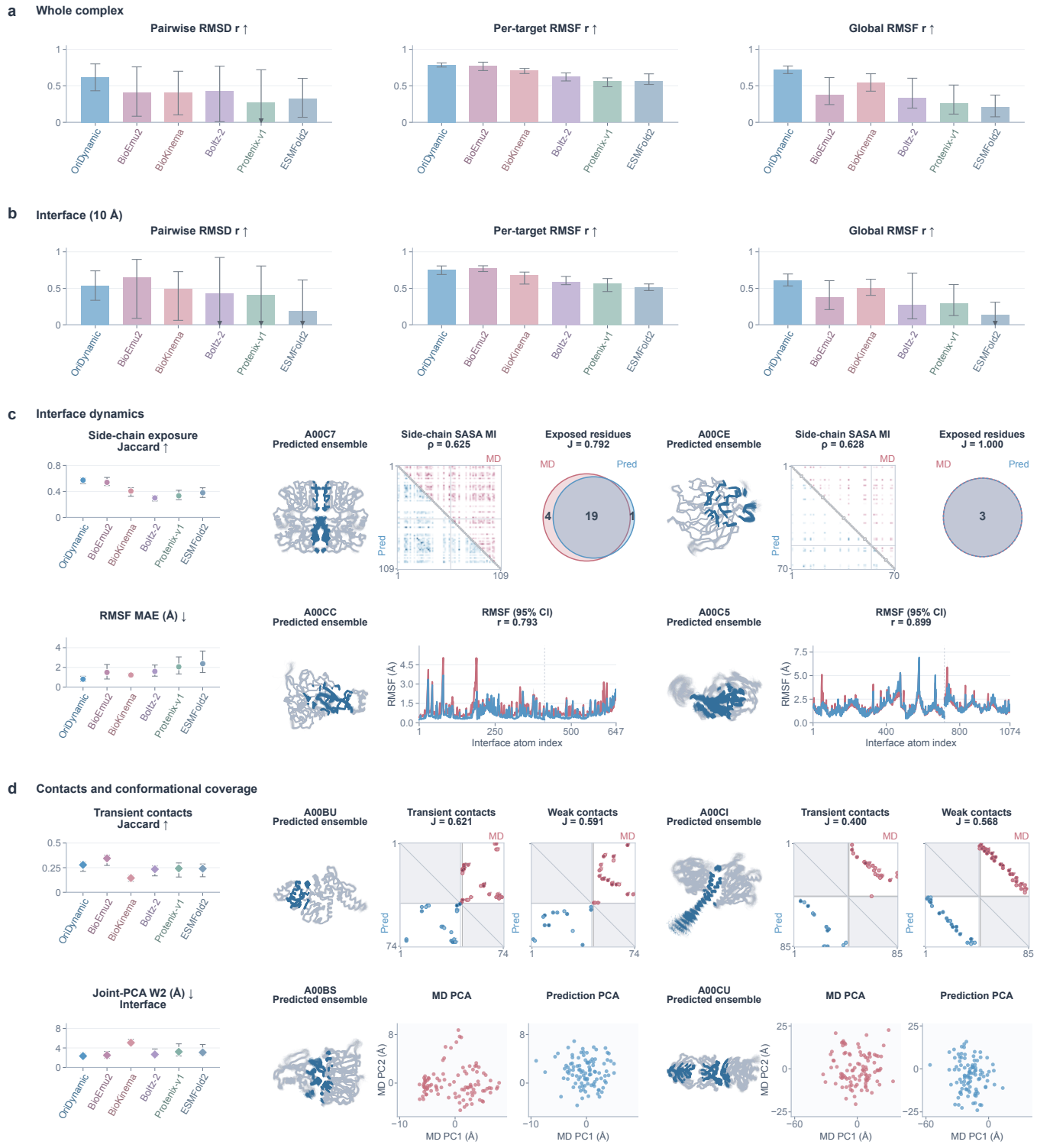


Figure 4. Protein–protein ensemble fidelity on DynaPPI. Six methods use 36 complexes, with 100 predictions and 100 MD frames per target. **a,b,** Whole-complex and 10 Å interface ensemble correlations; downward triangles indicate intervals extending below zero. **c,** Interface exposure and RMSF errors, with exposure-information maps and fluctuation profiles for selected cases. **d,** Contact recovery and conformational coverage, illustrated by contact maps and shared MD-PCA projections. MD is pink and predictions blue; matrix upper/lower triangles show MD/prediction. Summary bars are 95% target-bootstrap intervals and profile bands are pointwise frame-bootstrap intervals. RMSF correlations use heavy atoms; the primary amplitude-error, contact and PCA metrics retain their established Cα definitions. Cases were selected post hoc. Full protocols are in Appendix C.4.

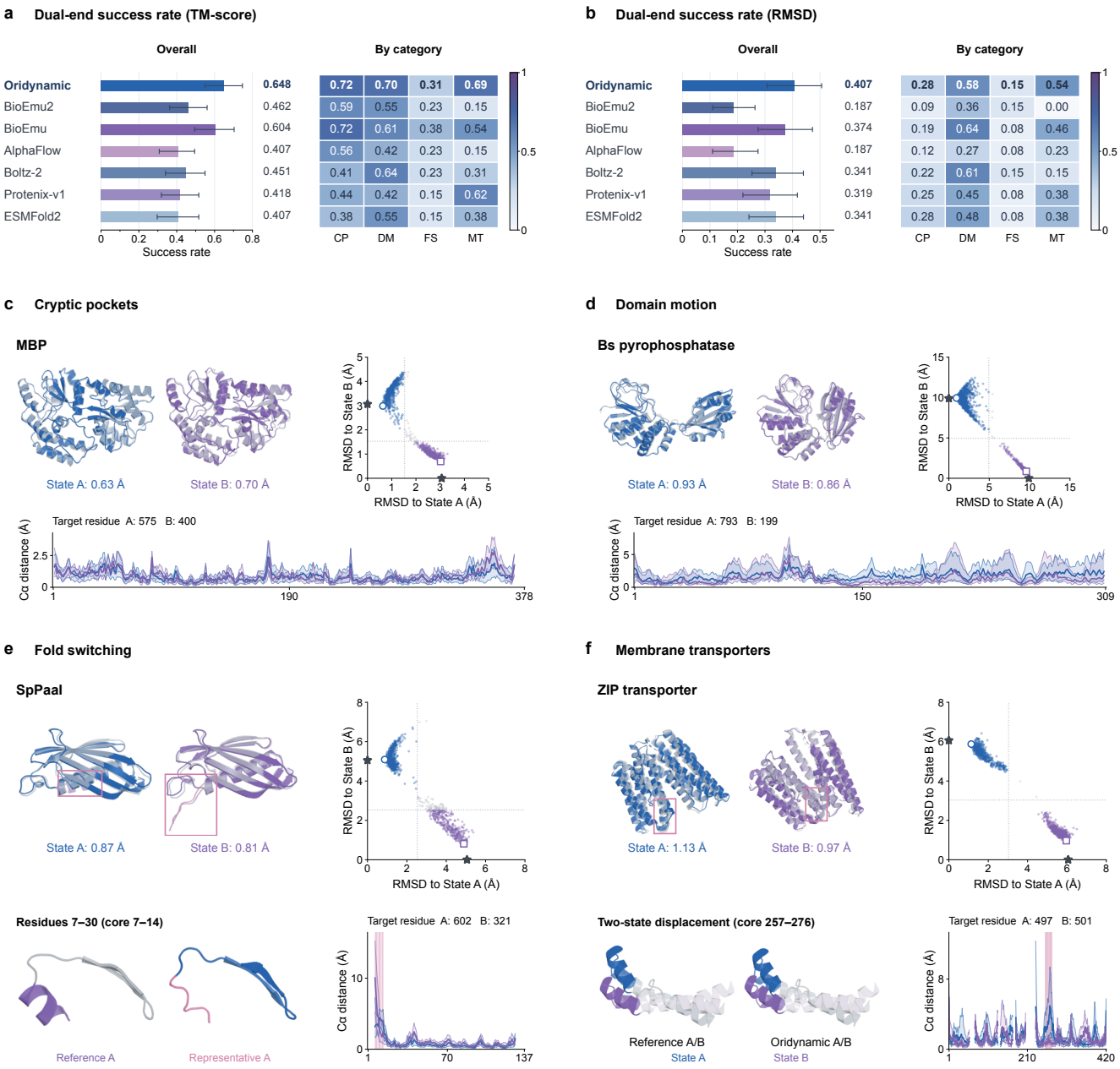


Figure 5. Recovery of alternative experimental conformations. a,b, Dual-end TM-score and RMSD success on 91 ConforMix targets, using 100 structures per method; bars show 95% target-bootstrap intervals. Matrices separate cryptic pockets, domain motion, fold switching and membrane transporters. **c–f,** MBP, Bs pyrophosphatase, SpPaaI and ZIP-transporter examples, each from a separate 1,000-structure OriDynamic ensemble. Blue/purple indicate reference-associated groups; gray indicates neither. Stars mark references, and outlined symbols mark minimum-RMSD representatives. Profiles show group-conditioned median Cα deviations and 10th–90th percentiles. Close-ups show the SpPaaI helix and ZIP-transporter displacement using the global alignment. Definitions and sampling settings are in Appendix C.5.

7/13 by RMSD. Category-specific comparisons showed that BioEmu recovered 5/13 fold-switching targets by TM-score and 21/33 domain-motion targets by RMSD, compared with 4/13 and 19/33 for OriDynamic.

We next examined four performance-selected cases, one per category, using separate ensembles of 1,000 structures generated with one final sampling seed. These ensembles were not included in the 100-structure benchmark. For maltose-binding protein, minimum RMSDs to States A and B were 0.63 and 0.70 Å, respectively (Fig. 5c). Bs pyrophosphatase reached 0.93 and 0.86 Å, with 793 and 199 conformers assigned to the corresponding RMSD-defined groups (Fig. 5d). The resulting ensembles contained conformers close to both experimental reference states.

SpPaaI provided a case of fold switching with a local secondary-structure rearrangement (Fig. 5e). Minimum RMSDs were 0.87 and 0.81 Å, and simplified DSSP identified the complete reference-A helix pattern at residues 7–14 in 5/1,000 conformers. The globally closest reference-A representative retained part of this local helix pattern.

For ZIP transporter, minimum RMSDs were 1.13 and 0.97 Å, with 497 and 501 conformers in the A- and B-associated groups (Fig. 5f). Close-ups compare the displacement of residues 257–276 between the two references with that between the corresponding representatives. Keeping the global alignment and a shared camera, without local refitting, preserves this spatial difference in both views.

The residue-deviation profiles complement the global comparisons by showing group-conditioned medians and 10th–90th percentiles along the target sequence. Together, the global comparisons and residue profiles show how the sampled ensembles cover alternative experimental conformations and their associated local rearrangements.

3.2 OriDynamic resolves partner-dependent conformational states

We next examined whether partner-conditioned sampling distinguishes the unbound and bound conformations of the same protein. The ProMiSE cohort comprised 456 experimental apo–holo pairs: 338 protein–ligand and 118 protein–protein pairs across 160 task-specific clusters. We compared OriDynamic with Boltz-2, Protenix-v1, ESMFold2 and BioEmu2 using 100 predictions per input condition. Double-state success required each branch to contain a prediction with an own-reference TM-score >0.8 and greater similarity to its own than the opposite state.

OriDynamic recovered both states for 45.4% of pairs, compared with 49.6% for ESMFold2, 32.2% for Boltz-2, 31.6% for BioEmu2 and 25.9% for Protenix-v1. Equal weighting of clusters changed the ranking: OriDynamic had the highest success estimate (54.7%), followed by ESMFold2 (46.9%), BioEmu2 (45.4%), Protenix-v1 (43.5%) and Boltz-2 (39.0%; Fig. 6a). OriDynamic also had the highest pair-weighted success on protein–protein pairs (46.6%), whereas ESMFold2 led on protein–ligand pairs (53.6% versus 45.0% for OriDynamic). Matched outcomes identified complementary apo recovery: OriDynamic recovered both states for 105 and 85 pairs on which Boltz-2 and Protenix-v1, respectively, recovered only the holo state (Fig. 6b,c).

The cluster-removal analysis further showed the influence of cohort composition. OriDynamic retained a positive success difference relative to Boltz-2, Protenix-v1 and BioEmu2 throughout removal of the 0–10 largest clusters; its difference from ESMFold2 became positive after the two largest clusters were removed (Fig. 6e, top). Uniform subsampling of the fixed prediction pools gave an expected success rate of 26.8% at five samples per branch, compared with 27.5% for ESMFold2, 21.1% for BioEmu2, 17.8% for Boltz-2 and 17.1% for Protenix-v1 (Fig. 6e, bottom).

Stratification by experimental apo–holo displacement separated endpoint recovery from the accuracy of the structural response (Fig. 6d). OriDynamic had the highest cluster-weighted success estimates in the low-, intermediate- and high-displacement strata (80.5%, 58.3% and 25.9%). Its residue-response Spearman correlations decreased from 0.616 to 0.556 and 0.466, while response-localization AP20 decreased from 0.614 in the low stratum to 0.544 in the high stratum. ESMFold2 had higher correlations in the intermediate and high strata (0.577 and 0.475) and higher high-stratum AP20 (0.570). OriDynamic's high-stratum direction cosine was also lower (0.184 versus 0.271 for ESMFold2). These measurements connect state recovery with the magnitude, localization and direction of the predicted structural response.

Selected protein–ligand examples illustrate the corresponding structural distributions (Fig. 6f). For the xylose-binding protein, the lowest-RMSD representatives among 100 samples per branch reached Cα RMSDs of 0.71 Å without xylose and 0.61 Å with xylose to their respective references. CMP kinase reached 1.01 and 0.93 Å without and with ara-CMP. The residue-wise profiles summarize medians and 10th–90th percentiles across all branch samples, showing local deviations alongside these best-of-100 representatives.

Protein partners provided complementary illustrations (Fig. 6g). LotA OTU1 with K6-linked diubiquitin and RimM with ribosomal protein S19 reached best-of-100 apo/holo RMSDs of 1.88/1.14 Å and 1.39/0.97 Å,

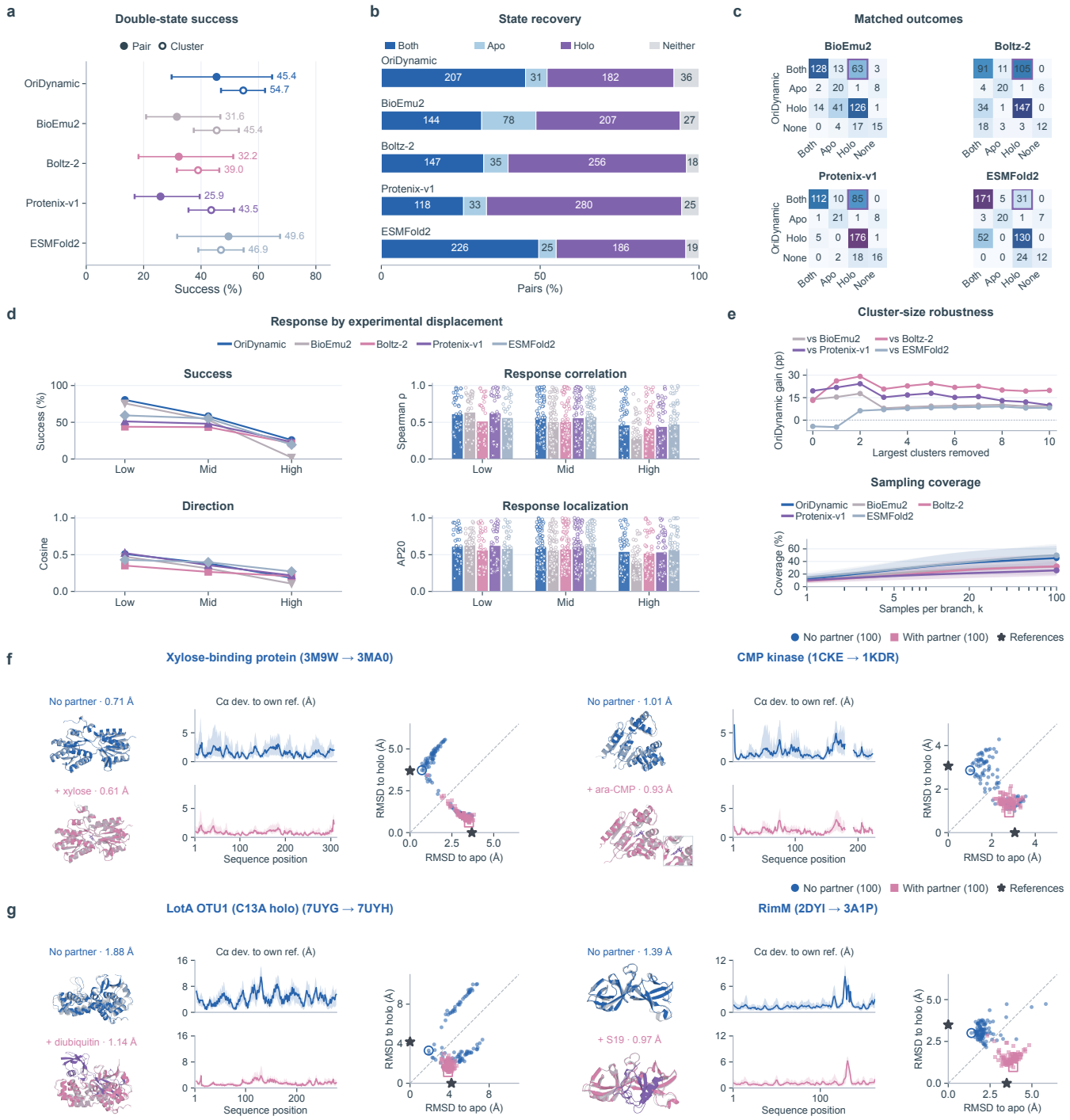


Figure 6. Partner-dependent conformational responses on ProMiSE. Five methods are evaluated on 456 apo–holo pairs (338 protein–ligand and 118 protein–protein), with 100 predictions per input condition. **a--c,** Dual-state success, outcome composition and paired method comparisons. **d,** Success and residue-level responses stratified by experimental displacement. Spearman plots omit 53 negative points from view; these remain in all means. **e,** Sensitivity to cluster removal and sampling budget. **f,g,** Selected protein–ligand and protein–protein cases, with best-of-100 representatives and per-residue median deviations with 10th–90th percentiles. Blue/pink indicate without-/with-partner predictions. Error bars and shaded uncertainty bands in a and e are 95% dependency-block bootstrap intervals. Reference policies, response definitions and display conventions are described in Appendix C.6.

respectively; the LotA holo reference contains the C13A substitution. These selected cases illustrate access to distinct experimental endpoints under different partner inputs. Across the ProMiSE benchmark, OriDynamic achieved the highest cluster-weighted dual-state success rate, with case ensembles showing distinct structural responses to ligand and protein partners.

3.3 OriDynamic captures ligand-pose distributions and interaction variability

We next assessed ligand conformational sampling on DD-13M, using the adapted OriDynamic model to generate 300 conformations for each of 28 test complexes. BioKinema and UnbindingFlow were evaluated with the same sample count and weighted MD reference ensembles. The analyses measured ligand-pose coverage, state-specific recovery, protein–ligand interactions and agreement in conformational distributions (Fig. 7).

At a 2 Å pose-RMSD threshold, OriDynamic achieved 66.0% mean coverage, compared with 52.3% for BioKinema and 40.6% for UnbindingFlow. Precision was 66.3%, compared with 95.5% and 9.2%, respectively. At 5 Å, OriDynamic's coverage and precision reached 94.3% and 97.9% (Fig. 7a). Thus, OriDynamic covered a larger fraction of the reference poses at the tighter threshold, while BioKinema retained the highest precision.

State-resolved analysis localized the coverage differences across the MD-defined S0–S3 states. OriDynamic achieved coverage of 96.4%, 81.2%, 52.4% and 13.3%, respectively, at 2 Å. For S2, coverage was 7.6% for BioKinema and 18.3% for UnbindingFlow; for S3, the corresponding values were 0.1% and 6.3% (Fig. 7b). OriDynamic therefore sampled reference-associated configurations across all four states, with the largest coverage advantage over BioKinema in S2.

Interaction-based metrics showed closer agreement with MD in contact occupancy and ligand displacement (Fig. 7c). Mean contact-probability MAE was 0.0207 for OriDynamic, compared with 0.0271 for BioKinema and 0.0400 for UnbindingFlow. Wasserstein-1 errors for the native-contact fraction were 0.201, 0.307 and 0.411, respectively; the corresponding ligand-centroid displacement errors were 1.43, 2.35 and 5.58 Å. These complementary measures relate pose-distribution agreement to changes in protein–ligand interactions.

Ligand RMSF correlation averaged 0.467 for OriDynamic, compared with 0.341 and 0.248 for BioKinema and UnbindingFlow (Fig. 7d). In the 4EEV and 2GQG examples, correlations were 0.851 and 0.844, with RMSF MAEs of 0.21 and 0.67 Å and predicted-to-MD amplitude ratios of 0.96 and 0.83 (Fig. 7e). For 1T31, pocket solvent-exposure mutual information correlated with MD at $r = 0.48$. The 4GU6 ligand44 example gave a contact-probability MAE of 0.092 over eight MD-selected variable contacts, compared with 0.113 across all 21 variable contacts (Fig. 7f).

Conformational-distribution analysis provided a further comparison in a fixed MD-derived ligand PCA space. Mean Wasserstein-2 distance was 2.78 Å for OriDynamic, versus 3.57 Å for BioKinema and 6.34 Å for UnbindingFlow, with lower errors on 23/28 and 27/28 complexes, respectively (Fig. 7g). For the distinct ligand39 and ligand45 complexes of 4GU6, the nearest sampled poses matched the four MD state medoids with worst-state RMSDs of 1.18 and 1.30 Å; their ensemble W_2 distances were 3.58 and 2.32 Å (Fig. 7h,j). Together, these DD-13M results show broader ligand-pose coverage and closer agreement with MD interaction and conformational distributions than the two evaluated baselines.

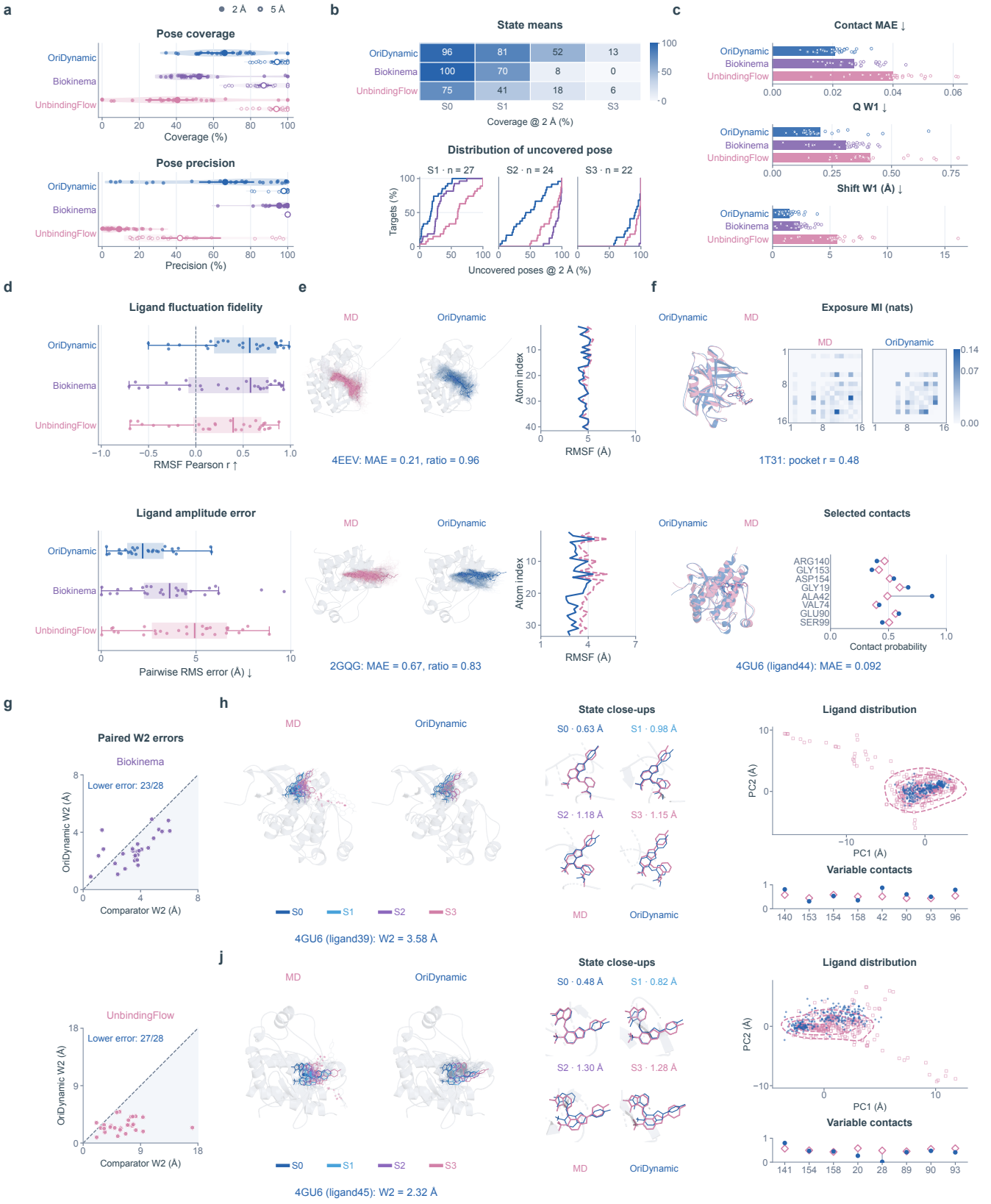


Figure 7. Ligand ensembles on DD-13M. Three methods use the same 28 complexes and 300 predictions per complex. **a,b,** Pose coverage/precision and MD-defined state coverage. **c,d,** Contact, displacement and ligand-fluctuation errors. Dots denote complexes; uncertainty bars are descriptive 95% sequence-component bootstrap intervals. **e,f,** Selected RMSF, pocket exposure-information and contact comparisons. **g,** Paired ligand-PCA W_2 errors. **h,j,** 4GU6 ligands39/45, showing MD state representatives, nearest predictions and ensemble distributions. MD is pink and OriDynamic blue in paired views; state colors link representatives and predictions. Cases were selected post hoc. All predictions enter the distribution metrics. Definitions and reference-frame counts are in Appendix C.7.

4 Methods

4.1 Data

We assembled the molecular dynamics (MD) corpus for joint training of OriDynamic from three complementary sources: ATLAS for protein monomers [27], MISATO for protein–ligand complexes [29], and mdCATH for protein domains [28]. After preprocessing and redundancy filtering, the training and validation sets contained 7,260 and 237 targets, respectively. Each dataset underwent target-level deduplication, and complete molecular targets were kept separate across training, validation, and test sets. Each target linked a molecular system and a structural reference to an ensemble of MD conformations. Training sampled the three sources with equal probability and then selected targets uniformly within each source. Conformations provided ensemble supervision without requiring temporal ordering. The MD test sets comprised 82 ATLAS proteins, 255 MISATO complexes, and 100 mdCATH domains.

Additional evaluations assessed conformational changes using 91 ConforMix targets [31], induced fit using 456 ProMiSE apo–holo pairs (338 protein–ligand and 118 protein–protein pairs) [32], and protein–protein ensemble fidelity using 36 DynaPPI complexes. For ligand dissociation, we adapted the joint model on 144 DD-13M systems, retained 30 systems for validation, and evaluated the adapted model on 28 systems [33]. The final ConforMix, ProMiSE, and DD-13M evaluation sets shared no complete molecular targets with the MD training set. Details of dataset preparation, partitioning, and redundancy control are provided in Appendix A.

4.2 Model Architecture

OriDynamic implements latent ensemble decomposition on an AlphaFold 3–like cofolding architecture. A molecular trunk provides context to three modules: an autoencoder (AE), a flow module, and a refine module. The input module embeds molecular composition and chemical features. The multiple sequence alignment (MSA) module incorporates protein evolutionary information into pair features, while the Pairformer updates single and pair representations to model relationships among molecular components. These representations condition structural encoding, latent generation, and all-atom sampling.

The AE learns a continuous latent representation through structural encoding and reconstruction. The flow module transforms Gaussian noise into latent states that decode into representative conformations across basins. Conditioned on these decoded representatives, the refine module performs stochastic all-atom sampling to explore fluctuations within each basin while improving local geometry. This generative decomposition represents conformational variability through alternative structural states and distributions of atomic configurations conditioned on them. Repeated sampling produces time-independent ensembles conditioned on molecular context.

4.3 Training and Loss Functions

Training follows three stages. First, we pretrain the AE using masked coordinate mean-squared error (MSE) and pair-distance reconstruction losses. Second, we freeze the AE and train the flow module by flow matching, learning a velocity field that transports Gaussian noise toward encoded MD conformations. Third, we introduce low-rank adaptation (LoRA) into the MSA and Pairformer modules and jointly optimize these adapters with the flow and refine modules, keeping the AE frozen. The AE retains fixed trunk context, while generation and refinement use adapted context. Ensemble gradients propagate through the refine module and frozen decoder to the flow module.

Statistical supervision complements the generative decomposition by constraining mean positions, variances, and joint covariance of the final generated ensemble:

$$\mathcal{L}_{\text{joint}} = \lambda_\mu \mathcal{L}_{\text{mean}} + \lambda_v \mathcal{L}_{\text{var}} + \lambda_c \mathcal{L}_{\text{cov}} + \mathcal{L}_{\text{aux}}. \tag{1}$$

The mean objective matches average atomic positions by MSE. The variance objective matches fluctuation amplitudes through root-mean-square fluctuations. The covariance objective constrains the full, trace-normalized joint coordinate covariance of selected representative atoms, capturing both individual directional variability and correlated displacements between atoms. Its normalization emphasizes relative fluctuation structure, complementing amplitude supervision. Statistics use common valid atoms in a shared coordinate frame without one-to-one pairing of generated and MD conformations. Auxiliary objectives include flow matching, coordinate denoising with noise-conditioned MSE on real MD frames, ensemble coverage, geometric penalties, and preservation of representative structures. Detailed definitions are provided in Appendix B.

4.4 Evaluation protocols

We evaluated conformational variability, distributional agreement, interactions and reference-state recovery using matched atoms and benchmark-specific alignments. Mean pairwise RMSD measured ensemble breadth, while atom-wise RMSF measured flexibility. For aligned coordinates X_{ka}, with K conformations, n atoms and atom-wise mean μ_a, RMSF and root-mean-squared Wasserstein discrepancy (RMWD) were

$$\mathrm{RMSF}_a = \left[K^{-1} \sum_{k=1}^{K} \|X_{ka} - \mu_a\|_2^2 \right]^{1/2}, \qquad \mathrm{RMWD}(P,R) = \left[n^{-1} \sum_{a=1}^{n} W_2^2\left(\mathcal{N}_a^P, \mathcal{N}_a^R\right) \right]^{1/2},$$

where $\mathcal{N}_a^P$ and $\mathcal{N}_a^R$ are Gaussian approximations to predicted and reference atomic coordinates. Pearson correlations and amplitude ratios summarized fluctuation agreement; PCA-space W_2 and scalar-observable W_1 measured distributional differences. Contact-probability errors, Jaccard overlap and solvent-exposure mutual-information correlations assessed interactions.

For ConforMix and ProMiSE, dual-state success required recovery of both experimental states with TM-score > 0.8 and greater similarity to the matched than the opposite state; ConforMix additionally used an RMSD criterion of half the reference-state separation. ProMiSE responses were assessed by displacement-direction cosine, residue-displacement Spearman correlation and AP20 (average precision for the most displaced 20% of residues). For DD-13M, weighted pose coverage was $C_\delta = \sum_j w_j \mathbf{1}[\min_i d(P_i, R_j) \le \delta]$ and precision was $P_\delta = K^{-1} \sum_i \mathbf{1}[\min_j d(P_i, R_j) \le \delta]$, where w_j are normalized MD weights and d is symmetry-corrected ligand RMSD after protein alignment; thresholds were 2 and 5 Å. Comparisons used matched target sets and sampling budgets, with 95% bootstrap intervals over targets or dependency blocks. Benchmark-specific atom selections, reference frames and statistical procedures are detailed in Appendix C.

5 Conclusions and future directions

OriDynamic represents biomolecular ensembles through latent ensemble decomposition: a flow module samples representative conformations across basins, and a conditional refine module explores variability within them. Supervision of mean positions, fluctuation amplitudes and joint covariance connects this generative organization to ensemble statistics. Evaluations on protein monomers, protein–ligand complexes and protein–protein assemblies, together with the three application studies, show how this framework captures conformational variability and partner-dependent structural responses. These results provide a basis for using ensembles to formulate structural hypotheses in molecular discovery.

Several capabilities remain to be established before this foundation can support a more complete biomolecular world model. Its represented world must encompass a broader range of entities and interactions, including nucleic acids, ions, cofactors, and complex multicomponent assemblies. It must also account for the external environment: temperature, pH, ionic strength, solvent, and membrane context can alter conformational behavior, and their effects should be captured through explicit conditioning and validated responses. A further requirement is a unified capacity for time-agnostic and time-aware modeling. These formulations address complementary questions—what states may be accessible and how states evolve over time—and should coexist within the same model. Completing these capabilities would allow the framework to support state exploration, environmental response prediction, and temporal process modeling according to the scientific question being asked.

We argue that genuine biomolecular intelligence requires the integration of a biomolecular domain model with an LLM. The domain model provides representations and predictions grounded in biomolecular structure, interactions, and conformational behavior; the LLM connects these capabilities to scientific questions, contextual knowledge, and goal-directed reasoning. Under this view, LLM integration is an essential component of the biomolecular world-model framework. It enables research intent to be translated into model queries and candidate interventions, generated states to be interpreted alongside scientific evidence, and predictions to be developed into experimentally testable hypotheses. Through this integration, a biomolecular world model could participate in an iterative scientific workflow: formulating hypotheses, exploring their structural consequences, proposing experiments, and revising conclusions in response to new observations. The resulting intelligence would emerge from the coupling of domain-grounded generation with language-based reasoning and empirical feedback.

Experimental feedback on selected hypotheses and designs can identify gaps in the generated ensembles and guide further model refinement.

Contributions

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Supplementary Information

A Data sources and preparation

A.1 Molecular dynamics data

ATLAS. ATLAS provides atomistic simulations of protein monomers [27]. We used the designated protein chain for each target and maintained the correspondence between its initialization structure and MD conformations through residue and named-atom mappings. The initialization structure was stored separately from the trajectory sampling pool. The retained ATLAS subset comprised 1,256 training, 39 validation, and 82 test targets.

MISATO. MISATO provides MD conformations of protein–ligand complexes [29]. We used the processed subset containing one protein chain and one ligand per target, with atom identities mapped from the source HDF5 representation. Protein and ligand coordinates were treated as parts of the same molecular system. We used 4,064 complexes for training, 98 for validation, and 255 for testing.

mdCATH. mdCATH provides MD simulations of structurally diverse protein domains [28]. The modeled target was the represented domain, with residue and named-atom mappings maintained across its conformations. Longer protein sequences available as multiple-sequence-alignment context were not treated as additional modeled residues when comparing target identities. The retained subset comprised 1,940 training, 100 validation, and 100 test domains.

Source	Training	Validation	Test
ATLAS	1,256	39	82
MISATO	4,064	98	255
mdCATH	1,940	100	100

Table 1. Source-specific training, validation, and test set sizes. Counts refer to molecular targets, each associated with an ensemble of conformations.

A.2 Molecular correspondence and ensemble sampling

Each processed target retained its molecular identities, structural reference, atom correspondence, coordinate-validity masks, and MD conformations. Coordinate masks determined which atoms contributed to supervision; a valid simulation coordinate was distinguished from experimental observability. The source-specific mappings preserved molecular identity across the reference and ensemble frames.

Joint training selected ATLAS, MISATO, or mdCATH with probability 1/3, followed by uniform target selection within the chosen source and sampling from that target's conformations. Consequently, source contributions were not proportional to the number of available targets. All conformations associated with a target remained in the same data partition. The ensemble objectives used conformational samples and ensemble statistics without supervision on temporal ordering or transition times.

A.3 Internal partitions and target identity

Each dataset underwent source-specific preprocessing to remove duplicate targets and target-level redundancy. The resulting training, validation, and test sets were non-overlapping at the level of complete molecular targets. All conformations belonging to the same target were assigned to the same partition. Target identity was assessed for the complete molecular system; a shared protein sequence or individual component alone did not constitute a duplicate target.

A.4 Conformational-change and induced-fit evaluations

ConforMix. The processed ConforMix evaluation set contained 91 targets [31]: 32 cryptic-pocket, 33 domain-motion, 13 fold-switching, and 13 transporter-related cases. Paired experimental structures defined the reference conformations.

ProMiSE. The final ProMiSE evaluation set contained 456 apo–holo pairs, comprising 338 protein–ligand and 118 protein–protein cases [32]. All methods were evaluated on the same retained cohort. The final ConforMix, ProMiSE, and DD-13M evaluation sets shared no complete molecular targets with the MD training set.

A.5 Protein--protein ensemble evaluation

The DynaPPI benchmark contained 36 protein–protein test complexes, with 100 reference MD frames per target. It was used to assess ensemble variability in whole complexes and at protein–protein interfaces. DynaPPI was not a source in the joint MD training corpus.

A.6 DD-13M adaptation and evaluation

DD-13M provides protein–ligand dissociation trajectories [33]. We used 144 systems for model adaptation, 30 for validation, and 28 for final testing. The source partitions were constructed using sequence components defined by a 40% sequence-identity threshold and at least 80% coverage on both sides of an alignment; source PDB identifiers and sequence components were kept within a single partition.

All methods were evaluated on the same 28 test systems, using 300 predicted conformations per system and method. The fixed MD reference ensembles contained up to 300 frames: 26 systems had 300 frames, and the remaining two had 102 and 279 frames, respectively; their original reference weights were retained.

B Model architecture and training

B.1 Data and sampling

The jointly trained model uses ATLAS protein monomers, MISATO protein–ligand complexes and mdCATH protein domains. Dataset sizes, internal partitions and target-level redundancy controls are given in Appendix A and Table 1. The DynaPPI protein–protein cohort is used for evaluation. Ligand-dissociation experiments use a subsequent adaptation on DD-13M. Initialization structures are stored separately from the MD frame-sampling pool.

Atom correspondence is established through molecular, residue, and atom identities, with coordinates and validity masks kept in the same order. Protein atom37 representations are mapped to model atom arrays; ligand atoms retain explicit chemical identities. Coordinates are expressed in Å. MD-coordinate validity is distinguished from experimental observability: simulation-valid atoms can contribute supervision even when unresolved experimentally. Invalid coordinates remain masked.

Joint-training sampling selects ATLAS, MISATO or mdCATH with equal probability, then a target uniformly within the selected source before drawing MD frames. AE training uses a single frame per micro-step; flow training and joint training use multiple conformations from the selected system. All frames of a target record retain the same split. Source-specific preparation and sampling are described in Appendix A. Temporal ordering is not a prediction target: frames supply structural examples and empirical ensemble statistics rather than transition labels.

Coordinate preprocessing preserves the relative placement of molecular partners. Whole-system centering and rotation augmentation act jointly on all partners. Reconstruction compares coordinates in the same augmented frame; ensemble statistics use a shared structural pose and common valid atom support. Dataset-specific preparation and overlap controls follow Appendix A.

B.2 Molecular trunk and latent ensemble decomposition

The AlphaFold 3–like cofolding trunk embeds molecular identity and chemical features while retaining atom-to-token mappings. The MSA module incorporates protein evolutionary information into pair representations. The Pairformer updates pair features through triangle multiplication, triangle attention, and transition layers, and updates single features through attention with pair-derived biases and transition layers. For N molecular tokens, its outputs comprise single representations $s \in \mathbb{R}^{N\times 384}$ and pair representations $z \in \mathbb{R}^{N\times N\times 128}$, alongside atom-level features.

Latent ensemble decomposition organizes generation into representative conformations across basins and conditional sampling of configurations within them. With molecular context C and a decoded representative X_{rep}, the hierarchical distribution is

$$p(X \mid C) = \int p_{\text{refine}}(X \mid X_{\text{rep}}, C) p_{\text{rep}}(X_{\text{rep}} \mid C)\, \mathrm{d}X_{\text{rep}}. \tag{S1}$$

Here p_{rep} is induced by latent flow sampling and structural decoding. A representative conformation serves both as a structural state and as a condition for further variability. The two levels are learned together through statistics of the final generated ensemble. This formulation produces time-independent ensembles rather than trajectories.

B.3 Structural encoding and reconstruction

The AE is initialized from a pretrained coordinate diffusion architecture, with atom attention encoding, a token transformer, and atom attention decoding. A latent bottleneck separates the encoding and decoding transformer blocks. The AE is deterministic, without posterior sampling or KL regularization. It learns a continuous representation of structural variation under fixed molecular context C_0.

Let X denote a target conformation and B a coordinate carrier. Shared encoding branches produce features U_X and U_B:

$$h = W_\mu \,\mathrm{LN}(U_X - U_B) + b_\mu, \qquad \widehat{X} = B + \Delta_\psi(h, B, C_0). \tag{S2}$$

The decoder projects h to token features, combines them with U_B, and predicts an all-atom displacement. Atom-level skip features originate from the carrier branch. In a latent-only representation, $B = 0$ and conformation-specific information passes through h. A coordinate–latent parameterization instead uses representative-site coordinates S as the carrier, with $B_a = S_{\tau(a)}$ for atom-to-token mapping τ. The flow module then jointly generates (h, S), so both components contribute to each representative conformation.

Reconstruction uses coordinate and pair-distance objectives:

$$\mathcal{L}_{\mathrm{AE}} = \mathcal{L}_{\mathrm{coord}} + \lambda_{\mathrm{pair}}\mathcal{L}_{\mathrm{pair}}. \tag{S3}$$

Coordinate MSE is averaged over valid Cartesian components within each entity and then across entities. Pair-distance squared errors are balanced across entity-pair groups. Sparse atom pairs provide local geometric supervision. Prediction and target are compared in their common coordinate frame without an independent prediction-to-target rigid fit.

B.4 Flow module: representative conformations across basins

After AE pretraining, its parameters are frozen. Encoded MD conformations provide training examples for the flow module. Channelwise latent means and standard deviations are estimated from training data and retained for sampling. For normalized latents $u = (h - \mu_h)/\sigma_h$, flow matching uses

$$u_t = (1-t)\epsilon + tu, \qquad \mathcal{L}_{\mathrm{FM}} = \mathbb{E}\left[\|v_\phi(u_t, t \mid s, z, r) - (u - \epsilon)\|^2_{\mathrm{masked}}\right], \tag{S4}$$

where $\epsilon \sim \mathcal{N}(0, I)$, $t \sim \mathcal{U}(0, 1)$, and r denotes an optional structural reference condition. The interpolation variable is not physical time. Coordinate–latent representations additionally include scaled representative-site coordinates, with latent and coordinate errors normalized over their respective valid entries.

The flow transformer uses time-dependent adaptive normalization and zero-initialized residual gates. Single features provide tokenwise modulation, while pair features supply attention biases. When enabled, structural reference features provide additional conditioning and are subject to condition dropout; molecular context remains available. Sampling integrates the velocity field from Gaussian noise using Heun updates. Inverse normalization and decoding produce representative all-atom conformations across basins.

B.5 Refine module: within-basin sampling and joint training

Conditioned on a decoded representative, the refine module performs stochastic all-atom sampling to resolve within-basin fluctuations and improve local geometry. It uses an independently parameterized coordinate diffusion network. For representative X_{rep}, refinement begins with

$$x_{\sigma_0} = X_{\mathrm{rep}} - \bar{X}_{\mathrm{rep}} + \sigma_0 \epsilon_x, \qquad \epsilon_x \sim \mathcal{N}(0, I), \tag{S5}$$

where $\bar{X}_{\mathrm{rep}}$ is the masked centroid. A differentiable Euler update along a descending noise schedule is

$$x_{i+1} = x_i + (\sigma_{i+1} - \sigma_i)\frac{x_i - D_\omega(x_i, \sigma_i \mid C_\theta)}{\sigma_i}. \tag{S6}$$

The schedule terminates at zero noise, and the representative centroid is restored after sampling. Independent coordinate noise produces distinct configurations conditioned on the same representative; stochastic rollout can also inject noise at intermediate steps. Thus, variability arises from both latent selection of representative states and conditional all-atom sampling.

Joint training updates the flow module, refine module, and LoRA adapters in the MSA and Pairformer modules. The AE and its molecular context C_0 remain fixed; the flow and refine modules use adapted context C_θ. Freezing

decoder parameters preserves the representation while allowing gradients from the final ensemble to pass through decoding to the flow module.

B.6 Ensemble-level supervision

The statistical decomposition constrains mean positions, variances, and joint covariance of final generated conformations. Let $P \in \mathbb{R}^{K\times n\times 3}$ and $T \in \mathbb{R}^{M\times n\times 3}$ be generated and reference ensembles on the same valid representative atoms in a shared coordinate frame. Representative atoms are protein Cα atoms and, for PL systems, ligand heavy atoms. All partners undergo any alignment together, so intermolecular geometry is preserved. Generated and reference samples need not be paired one-to-one.

With atom weights $w_a > 0$ and $W = \sum_a w_a$, the mean and variance objectives are

$$\mathcal{L}_{\text{mean}} = \frac{1}{3W}\sum_a w_a \|\bar{P}_a - \bar{T}_a\|_2^2, \tag{S7}$$

$$V_a^P = \frac{1}{K}\sum_k \|P_{ka} - \bar{P}_a\|_2^2, \qquad V_a^T = \frac{1}{M}\sum_m \|T_{ma} - \bar{T}_a\|_2^2, \tag{S8}$$

$$\mathcal{L}_{\text{var}} = \frac{1}{W}\sum_a w_a \left(\sqrt{V_a^P + \varepsilon_v} - \sqrt{V_a^T + \varepsilon_v}\right)^2, \tag{S9}$$

with $\varepsilon_v = 0.01\,\text{Å}^2$. The variance objective compares root-mean-square fluctuation amplitudes. Protein-only systems use uniform atom weights; PL systems assign equal total weight to protein Cα atoms and ligand heavy atoms.

Define rows of F and G by flattening $\{\sqrt{w_a}(P_{ka} - \bar{P}_a)/\sqrt{K}\}_a$ and $\{\sqrt{w_a}(T_{ma} - \bar{T}_a)/\sqrt{M}\}_a$, respectively. Normalize each matrix by its Frobenius norm, with a numerical floor, to obtain $\widetilde{F}$ and $\widetilde{G}$. The joint covariance objective is

$$\mathcal{L}_{\text{cov}} = \|\widetilde{F}^\top\widetilde{F} - \widetilde{G}^\top\widetilde{G}\|_F^2 \tag{S10}$$

$$= \|\widetilde{F}\widetilde{F}^\top\|_F^2 + \|\widetilde{G}\widetilde{G}^\top\|_F^2 - 2\|\widetilde{F}\widetilde{G}^\top\|_F^2. \tag{S11}$$

Away from the numerical floor, this compares the full, trace-normalized joint coordinate covariance of the selected atoms. Diagonal 3×3 blocks describe individual directional variability, and off-diagonal blocks capture correlated displacements between atoms. Trace normalization emphasizes the relative structure of joint fluctuations, complementing variance-based amplitude supervision. The equivalent Gram expression avoids constructing dense $3n \times 3n$ covariance matrices.

Auxiliary supervision combines bidirectional nearest-neighbor ensemble coverage, bond-length and clash penalties, and preservation of decoded representative geometry through representative-atom pair distances. The preservation target is detached on the decoded-representative side. Real MD conformations also provide flow-matching targets and coordinate-denoising supervision, including noise-conditioned MSE and structural objectives:

$$\begin{aligned}\mathcal{L}_{\text{aux}} = \lambda_{\text{cover}}\mathcal{L}_{\text{cover}} + \lambda_b\mathcal{L}_{\text{bond}} + \lambda_{\text{clash}}\mathcal{L}_{\text{clash}} \\ + \lambda_p\mathcal{L}_{\text{preserve}} + \lambda_{\text{FM}}\mathcal{L}_{\text{FM}} + \lambda_{\text{fold}}\mathcal{L}_{\text{fold}}.\end{aligned} \tag{S12}$$

Together with the main-text joint objective, these terms connect representative state generation and within-basin sampling to collective statistics and local structural quality of the final ensemble.

B.7 Optimization and sampling settings

The jointly trained model follows the three-stage procedure described above. Optimization uses AdamW, gradient clipping, and mixed precision; the flow module maintains an exponential moving average of its parameters. Joint training uses separate learning rates for the flow module, refine module, and trunk adapters. Evaluation sample budgets and the checkpoint used for DD-13M adaptation are specified in Appendix C.

C Supplementary evaluation details

C.1 Shared definitions and statistical analysis

We evaluate conformational variability, distributional agreement, interaction recovery and coverage of reference states. The same definitions apply across the monomer, protein–ligand and protein–protein benchmarks; atom

selections, alignment masks, sampling budgets and aggregation are specified below for each task. Molecular identities establish atom correspondence before alignment. A single rigid transformation is applied to each complete complex, preserving the relative positions of its components. Distance-based ensemble metrics use the resulting common coordinate frame; comparisons with individual experimental states use the state-specific fits described below. Coordinates and coordinate-based distances are reported in Å.

Conformational breadth and fluctuations. For an aligned ensemble $X = \{X_k\}_{k=1}^{K}$ with n evaluated atoms, let $X_{ka} \in \mathbb{R}^3$ denote atom a in conformation k and $\mu_a = K^{-1} \sum_k X_{ka}$. Pairwise root-mean-square deviation and atom-wise root-mean-square fluctuation are

$$d(X_k, X_l) = \left[\frac{1}{n} \sum_{a=1}^{n} \|X_{ka} - X_{la}\|_2^2 \right]^{1/2}, \qquad \mathrm{RMSF}_a(X) = \left[\frac{1}{K} \sum_{k=1}^{K} \|X_{ka} - \mu_a\|_2^2 \right]^{1/2}. \tag{S13}$$

Mean pairwise RMSD averages d over the frame pairs selected by each benchmark, without additional pairwise fitting. The native all-pairs implementation averages the full prediction–prediction distance matrix, including its zero diagonal; reference ensembles use the fixed draws specified below. Pairwise-RMSD Pearson correlations compare target-level ensemble means. Global RMSF correlations pool matched atom values across targets, whereas per-target RMSF correlations compare profiles within each target and are summarized by their median. Pearson r is the centered dot product divided by the product of centered Euclidean norms. Amplitude ratios retain the benchmark's stated aggregation, distinguishing a ratio of cohort medians from a within-target RMS fluctuation ratio.

Gaussian Wasserstein discrepancy. Following the atomic Gaussian comparison used in AlphaFlow [21], we compute each atom's coordinate mean μ_a and population covariance $\Sigma_a = K^{-1} \sum_k (X_{ka} - \mu_a)(X_{ka} - \mu_a)^\top$. For predicted ($P$) and reference ($R$) ensembles,

$$g_a^2 = \|\mu_a^P - \mu_a^R\|_2^2 + \mathrm{tr}\left[\Sigma_a^P + \Sigma_a^R - 2 \left((\Sigma_a^R)^{1/2} \Sigma_a^P (\Sigma_a^R)^{1/2} \right)^{1/2} \right], \tag{S14}$$

$$\mathrm{RMWD}(P, R) = \left(\frac{1}{n} \sum_a g_a^2 \right)^{1/2}. \tag{S15}$$

For weighted DD-13M reference ensembles, means, covariances and RMSF use the retained normalized frame weights in place of uniform averages. The squared discrepancy separates into mean-position and covariance contributions. This evaluation compares the marginal 3×3 covariance of each atom; the training objective in Appendix B.6 instead compares the full joint covariance across selected atoms.

Distributional distances. For projected coordinate samples u_i and v_j with normalized weights p_i and q_j, the PCA Wasserstein-2 distance is

$$W_{2,\mathrm{PCA}}(P, R) = \left[\min_{\pi \in \Pi(p,q)} \sum_{i,j} \pi_{ij} \frac{\|u_i - v_j\|_2^2}{n} \right]^{1/2}, \qquad \Pi(p, q) = \{\pi \geq 0 : \pi \mathbf{1} = p, \pi^\top \mathbf{1} = q\}. \tag{S16}$$

The same PCA basis is used for both ensembles. For equally weighted, equal-sized samples, the monomer and protein–protein implementations solve a minimum-cost one-to-one assignment in the first two principal components. A joint basis is fitted to reference and predicted conformations together; case visualizations that use an MD-fitted basis are identified separately. DD-13M fits its PCA basis to the weighted MD ensemble and solves balanced discrete optimal transport with uniform prediction weights and retained reference weights. Its implementation applies the equivalent $1/\sqrt{n}$ coordinate normalization before PCA, so atom-count normalization is applied only once. For scalar observables, including interaction distances, native-contact fractions and centroid displacements, Wasserstein-1 distance is

$$W_1(P, R) = \int_{-\infty}^{\infty} |F_P(t) - F_R(t)| \, \mathrm{d}t, \tag{S17}$$

where F_P and F_R are the corresponding empirical cumulative distributions.

Contacts and solvent exposure. A contact probability is the ensemble-weighted frequency of a selected distance falling below its threshold. On a fixed contact set $\mathcal{C}$, probability error and binary-set overlap are

$$\mathrm{MAE}_{\mathrm{contact}} = \frac{1}{|\mathcal{C}|}\sum_{c\in\mathcal{C}} |p_c^P - p_c^R|, \qquad J(A,B) = \frac{|A\cap B|}{|A\cup B|}. \tag{S18}$$

The protein–protein implementation assigns zero when the union is empty. Binary solvent-exposure variables E_a, E_b yield mutual information $I(E_a;E_b) = \sum_{u,v\in\{0,1\}} p_{ab}(u,v)\log[p_{ab}(u,v)/(p_a(u)p_b(v))]$, with zero-probability terms omitted. Natural logarithms give units of nats. Spearman correlation compares the corresponding reference and predicted mutual-information entries.

Reference-state recovery and pose coverage. Dual-state success requires recovery of both experimental states under the task-specific TM-score or RMSD criterion; different samples can recover the two states. For a pose-distance threshold δ, reference weights w_j summing to one, reference poses R_j and K predictions P_i, coverage and precision are

$$\mathrm{Coverage}_\delta = \sum_j w_j\,\mathbf{1}\left[\min_i d(P_i,R_j)\le\delta\right], \qquad \mathrm{Precision}_\delta = \frac{1}{K}\sum_i \mathbf{1}\left[\min_j d(P_i,R_j)\le\delta\right]. \tag{S19}$$

DD-13M uses ligand heavy-atom RMSD in the shared protein-aligned frame, minimized over the stored ligand-atom symmetry permutations. State-specific coverage restricts reference poses to the corresponding MD-defined state and renormalizes their weights. For partner-dependent responses, the predicted displacement is the difference between mean with-partner and without-partner coordinates after alignment to apo. Its direction cosine with the experimental displacement, residue-displacement Spearman correlation and AP20 characterize direction, ranking and recovery of the most displaced residues, respectively. AP20 is average precision with the experimentally most displaced 20% of residues treated as positives.

Aggregation and uncertainty. Comparisons retain matched targets and reference data across methods. Target-level bootstrap resampling preserves all statistics belonging to a target; cluster or sequence-component resampling preserves the corresponding dependency blocks. Unless specified otherwise, intervals are the 2.5th and 97.5th percentiles of the bootstrap distribution. Each benchmark retains its original resampling unit and replicate count. Lower discrepancies and errors indicate closer agreement; higher correlations, overlaps and state-recovery scores indicate stronger correspondence.

C.2 Monomer ensemble evaluation

The evaluation in Fig. 2 uses 82 ATLAS and 100 mdCATH targets for every method, with 250 predictions per target. The evaluation cache contains 301 ATLAS reference conformations (including the initial structure) and the first 400 mdCATH reference frames. This reference-cache convention is distinct from the training frame-sampling pool. Cα atoms are selected before scoring, matched by residue identity and rigidly aligned to the fixed initial Cα reference. The same atom correspondence, reference ensembles and native metric implementation are used for all ten methods.

The original metric-specific reference sampling is retained. With pseudorandom seed 137, the evaluator draws two sets of 250 reference indices with replacement for the reference pairwise-RMSD statistic; all predicted pairwise distances are included. RMWD uses a further with-replacement reference draw of size $\min(1000, N_{\mathrm{MD}})$ and all 250 predicted structures. The metric is defined in Eqs. (S14)–(S15). Joint-PCA W_2 uses the first two components of a PCA basis fitted jointly to the first 250-frame reference draw and all predicted frames; assignment costs are normalized by the number of evaluated Cα atoms. RMSF uses the complete reference and prediction arrays. The saved reference draws are identical across methods for a given target.

Correlations follow Section C.1. Confidence intervals use 10,000 paired target-bootstrap resamples with seed 137. Secondary-structure and length-stratified comparisons retain the same per-target statistics.

OriDynamic uses joint11500_final_wo_v2. BioEmu2 uses equilibrium sampling. BioKinema uses five trajectories initialized from the reference MD starting structure, with 50 predicted frames per seed after excluding frame zero. The labels AlphaFlow and ESMFlow refer to the evaluated MD variants. mdCATH AlphaFlow uses the complete historical 100-target batch. ATLAS Protenix-v1 comprises 68 historical targets and 14 completed targets; the latter used verified MSAs and no effective templates. Evaluation uses matched atom coverage and ensemble sizes with these method-specific generation protocols.

Secondary-structure grouping uses only MD references: DSSP assignments are averaged over all reference frames and evaluated residues, with the same descriptive thresholds for both datasets. No predicted performance enters the grouping. Length bins are < 100, 100–199 and ≥ 200 evaluated residues. Case statistics retain all stored conformations; subsampling is used only for background structure traces.

C.3 Protein--ligand ensemble evaluation

Figure 3 uses the shared metrics in Section C.1 with the following settings.

OriDynamic, BioEmu2, BioKinema, Boltz-2, Protenix-v1 and ESMFold2 are evaluated on the same 255 targets, using 100 saved structures per method and target, 100 MD reference frames (0–99), and matched atom identities. BioEmu2 uses equilibrium sampling.

(a–c) Ligand-heavy-atom metrics: (a) across-target Pearson correlation of mean pairwise RMSD; (b) median of within-target atom-wise RMSF Pearson correlations; (c) global Pearson correlation after concatenating paired ligand-atom RMSF values from all targets (5,708 matched atoms), retaining each ligand's atom count. Pairwise RMSD uses all prediction pairs and two fixed, resampled MD frame sets, without pairwise refitting. Error bars in a–c and e denote percentile 95% confidence intervals (CIs) from 2,000 paired target-bootstrap resamples; global RMSF resamples whole target blocks.

(d) Wasserstein (WS) distances between reference-defined heavy-atom interaction-distance distributions, averaged over features within each target and category. Bars show target medians, thick intervals the interquartile range, thin whiskers the observed range, and dots individual targets. Each category uses the same eligible target subset across methods; n is indicated above.

(e) Ligand pairwise-spread and RMSF-amplitude ratios compare separately computed across-target medians (prediction/MD; dashed line, unity). Pocket pairwise RMSD agreement is the across-target Pearson correlation; pocket RMSF-pattern agreement is the median within-target atom-wise Pearson correlation. The pocket comprises ligand atoms and protein heavy atoms within 10 Å of the ligand in MD frame 0. All regional metrics use full-complex shared-heavy-atom alignment to that frame before selecting ligand or pocket atoms.

(f) Predicted versus MD pocket pairwise RMSD (upper) and mean pocket RMSF (lower), with one point per target. Lines and shaded bands show ordinary least-squares fits and 95% paired-bootstrap mean-fit CIs, calculated in Å before display transformation; dashed lines indicate equality. Annotated r values are across-target Pearson correlations.

(h,i) Illustrative ensembles for 2XZG and 5L9L, showing all 100 MD and OriDynamic structures. MD is pink, OriDynamic blue, and ligands orange. Pocket RMSF curves include 95% conditional sampling intervals from 2,000 resamples (whole conformers for predictions; circular blocks of 10 frames for MD); gold marks identify ligand atoms. PCA is fitted to MD pocket coordinates only; both ensembles use the same basis and MD-derived density bandwidth. Contours enclose 50% and 80% of evaluated density-grid mass; axis percentages report MD variance explained. The two cases were selected for favorable agreement with MD.

C.4 Protein--protein ensemble evaluation

Figure 4 uses the shared definitions in Section C.1 with the following task-specific settings.

OriDynamic, BioEmu2, BioKinema, Boltz-2, Protenix-v1 and ESMFold2 are evaluated on the same 36 complexes, with 100 predicted conformations and 100 reference MD frames per target. BioEmu2 denotes equilibrium sampling.

(a,b) Whole-complex and 10 Å interface comparisons. Pairwise RMSD reports the cross-target Pearson correlation of mean ensemble pairwise RMSD. Per-target RMSF reports the median within-target Pearson correlation; global RMSF reports the Pearson correlation of concatenated profiles.

(c) Interface solvent exposure and fluctuation amplitude. Summary metrics are the median exposed-residue Jaccard score and the mean interface RMSF absolute error. A00C7 and A00CE show predicted ensembles, side-chain solvent-exposure mutual-information maps and exposed-residue overlaps; ρ denotes the Spearman correlation of mutual-information values. A00CC and A00C5 show interface RMSF profiles with pointwise 95% confidence intervals; r denotes Pearson correlation. The horizontal axis indexes interface atoms in their original order, with chain boundaries marked.

(d) Interface contact recovery and conformational coverage. Summary metrics are the median transient-contact Jaccard score and median joint-PCA Wasserstein-2 distance. A00BU and A00CI show transient and weak contact maps. Contacts use an 8 Å threshold; transient contacts are absent from the input structure and populated with probability greater than 0.1, whereas weak contacts are present in the input structure and populated with

probability below 0.9. A00BS and A00CU show projections into the same first two components fitted to each reference MD interface ensemble. The case projections use an MD-fitted basis, and the summary Wasserstein-2 distances use a joint basis.

MD is shown in pink and predictions in blue. Matrix upper triangles show MD and lower triangles show predictions, with common scales within each matrix type. Mutual information is measured in nats and contact entries represent probabilities. Structure views contain all 100 predicted traces, with interfaces highlighted. All eight cases are distinct and were selected post hoc for illustration.

Summary error bars are 95% percentile intervals from 10,000 target-bootstrap resamples. RMSF profile bands use 10,000 frame-bootstrap resamples without realignment and do not account for temporal dependence. RMSF correlations and case profiles use heavy atoms; pairwise RMSD, the primary RMSF absolute-error metric, contacts and PCA retain the established Cα definitions. Solvent-exposure analyses use side-chain SASA. Method-specific matching and alignment follow the original evaluation protocols.

C.5 Alternative-conformation evaluation

Figure 5 uses the shared definitions in Section C.1 with the following task-specific settings.

a,b, Seven methods are compared on 91 ConforMix targets, with 100 evaluated structures per target and method. OriDynamic denotes no_anchor/Joint11500; BioEmu2 uses equilibrium sampling. Bars show dual-end success fractions with 95% percentile intervals from 1,000 target bootstrap replicates. Matrices show cryptic pockets (CP, n=32), domain motion (DM, n=33), fold switching (FS, n=13) and membrane transporters (MT, n=13), on a common 0–1 scale. TM-score success requires a sample for each state with TM-score >0.8 and higher than its opposite-state score. RMSD success requires both state-specific minima to be ≤half the reference-to-reference RMSD. USalign TM-scores use fixed residue correspondence and full target-length normalization. All 100 outputs per target are evaluated without quality filtering; sample counts are matched, whereas model inputs and computational costs differ.

c–f, Four performance-selected OriDynamic cases: MBP (c), Bs pyrophosphatase (d), SpPaaI (e) and ZIP transporter (f). Each uses a separate 1,000-structure ensemble generated with one final sampling seed and Joint11500 weights; these ensembles do not enter a,b. Global views show gray references and blue/purple minimum-RMSD representatives for A/B, with RMSDs labeled. Scatters retain all 1,000 samples: blue/purple denotes RMSD ≤half the reference separation and closer to A/B; gray denotes neither. Stars mark references, dashed lines thresholds, and outlined circles/squares the A/B representatives. Profiles show median Cα deviations from the corresponding reference and 10–90% intervals within each color group after global shared-Cα alignment; labels give group sizes and gaps indicate missing residues. The intervals summarize within-group conformational variation; group sizes describe the generated samples. Pink boxes and profile bands mark local cores. In e, purple/pink highlights residues 7–14 in reference A/representative A within residues 7–30. Simplified DSSP detects the full reference helix pattern in 5/1,000 samples; the displayed A representative lacks that complete pattern. In f, reference A/B conformations are overlaid at left and representatives at right; blue/purple marks A/B residues 257–276, with residues 238–298 as pale context. This core maximizes mean reference displacement among internal 20-residue windows. Paired close-ups share the global alignment, camera and scale, without local refitting.

C.6 Partner-dependent response evaluation

Figure 6 uses the shared definitions in Section C.1 with the following task-specific settings.

Evaluation on 456 ProMiSE apo–holo pairs (338 protein–ligand, 118 protein–protein; 160 task-specific clusters). Each method contributes 100 predictions without and 100 with the partner per pair. Double-state success requires each branch to contain a sample with own-reference-normalized USalign TM-score >0.8 and greater similarity to its own than the opposite state.

a, Pair-weighted (filled circles) and cluster-weighted (open circles) double-state success. b, State-recovery composition; labels give pair counts, and Apo/Holo denote apo-only/holo-only. c, Matched outcomes with OriDynamic in rows and each named baseline in columns; outlined cells identify both-state recovery by OriDynamic versus holo-only recovery by the baseline.

d, Responses stratified by apo-fitted experimental displacement, using fixed cutoffs of 4.1925280879 and 9.1781703606 Å: 151/149/156 pairs and 41/82/50 within-stratum clusters in the low/mid/high groups. Clusters can span strata. Lines show equal-cluster means for success and response-direction cosine; bars and open circles show equal-cluster means and individual cluster means for Spearman correlation and AP20. Predicted response is the difference between the mean with-partner and without-partner coordinates after independent alignment

to apo on the common Cα mask. Direction is its cosine with the experimental displacement (reference RMSD $\geq$0.5 Å); Spearman compares per-residue displacement-magnitude ranks; AP20 measures average precision for identifying the 20% most displaced experimental residues. All cluster values, including negative correlations, enter the reported means.

e, Pair-weighted OriDynamic success minus each baseline after removing the 0–10 largest clusters (top), and expected double-state coverage under uniform subsampling without replacement from each fixed 100-model branch (bottom). Error bars in a and shaded bands in e are pointwise 95% percentile intervals from 5,000 paired dependency-block bootstrap replicates; removal curves have no intervals.

f,g, Selected OriDynamic protein–ligand and protein–protein examples. Blue circles and pink squares identify the without-partner and with-partner branches, respectively; each case contains 100 predictions per branch. Scatter coordinates are Cα RMSDs to apo and holo after independent fits; the dashed line marks equal RMSD, stars mark experimental references, and open symbols mark the minimum own-reference-RMSD representatives shown as structures. Gray structures are references and purple structures are partners. Residue profiles show median own-reference Cα deviations and 10th–90th percentiles across all branch samples, with missing positions retained as gaps; these are not RMSF. Partners retain the target-derived alignment. LotA's holo reference contains C13A.

Apo predictions use best-observed apo references for TM scoring. Holo-versus-apo scoring uses pair-specific apo references for OriDynamic, ESMFold2 and BioEmu2, and best-observed apo references for Boltz-2 and Protenix-v1; response and case geometry use pair-specific references. The comparison uses equal sample counts with method-specific generation protocols. Response metrics are calculated on the target protein chain.

C.7 Ligand-dissociation ensemble evaluation

Figure 7 uses the shared definitions in Section C.1 with the following task-specific settings.

All methods use 300 predictions for each of 28 complexes. For normalized frame weights w_k, the pairwise RMS amplitude is $A(X) = \{2\sum_a \mathrm{RMSF}_a(X)^2/[n(1-\sum_k w_k^2)]\}^{1/2}$; its reported error is $|A(P) - A(R)|$. This weighted root-mean-square amplitude is distinct from the mean pairwise RMSD used in the other ensemble benchmarks. Weighted C300 MD references contain 300 frames for 26 complexes and 102/279 frames for the other two. OriDynamic uses validation-selected Joint11500 step1500 with true-init codec anchoring; explicit denoiser reference conditioning and supervised MD coordinates are absent during generation. OriDynamic is blue, BioKinema purple, UnbindingFlow pink; MD is pink in paired views. a, Pose coverage/precision at 2/5 Å: dots denote complexes, large markers show means, and intervals are descriptive 95% sequence-component bootstrap intervals. b, State coverage and uncovered-pose distributions; S0–S3 support 28/27/24/22 complexes. c, Contact-probability MAE, native-contact fraction Q W1 and centroid-displacement W1. d, Ligand RMSF correlation and pairwise RMS amplitude error; boxes show quartiles/median and dots show complexes. e, 4EEV/2GQG ensembles and unscaled atom-wise RMSF; ratio denotes predicted/MD root-mean-square RMSF. f, 1T31 weighted binary sidechain-exposure MI for 16 MD-defined pocket residues, with a shared square-root scale; 4GU6 ligand44 contacts use a 4.5 Å threshold and eight highest-variance MD contacts among 21 variable contacts (selected/all-variable-contact MAE 0.092/0.113). g, Paired ligand W2. h,j, 4GU6 ligands39/45: ensemble locations, MD state medoids and nearest prediction matches, original MD-PCA distributions and MD-selected contacts. State colors mark MD representatives and matched predictions. Ensemble pairs share cameras/scales and complete X0 protein contexts; close-ups retain every ligand atom. Cases were selected post hoc. All predictions enter the distribution metrics.

C.8 Ablation studies

We examine two aspects of the framework: the contribution of the refinement module within OriDynamic, and the effect of the structural-anchor source. Both comparisons cover ATLAS, mdCATH, MISATO and DynaRepo, using the reported correlations for pairwise RMSD, Global RMSF and Per-target RMSF. MISATO scores are evaluated on ligand atoms only, whereas DynaRepo scores are evaluated at the complex level.

C.8.1 Contribution of structural refinement

We compare AE+FM, the decoded conformations generated by OriDynamic's latent flow-matching module, with AE+FM+RM, the conformations obtained after applying the diffusion refinement module. Both configurations use PTX-predicted structural anchors. This comparison assesses the contribution of refinement with the latent generator retained; it does not separately ablate the autoencoder or the flow-matching module.

As shown in Fig. 8, refinement increases all three reported correlations on ATLAS, mdCATH and DynaRepo. On ATLAS, the scores change from 0.710/0.778/0.842 to 0.834/0.833/0.892, in the order of pairwise RMSD,

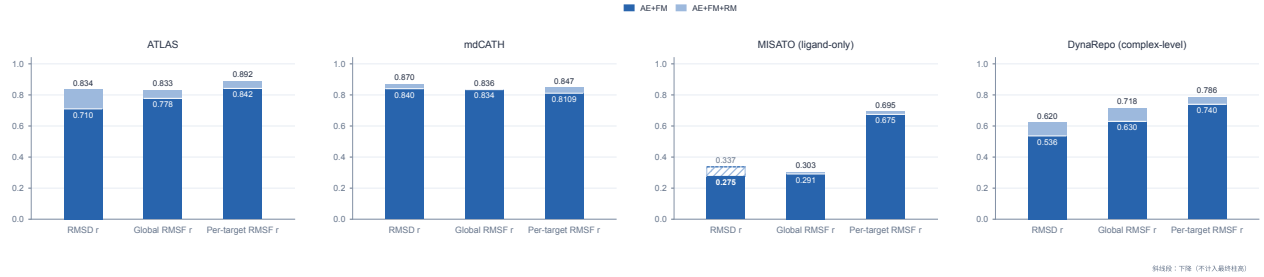


Figure 8. Contribution of the refinement module to OriDynamic. The four panels, from left to right, show ATLAS, mdCATH, MISATO (ligand-only), and DynaRepo (complex-level). Each panel reports pairwise RMSD, Global RMSF, and Per-target RMSF correlations for AE+FM and AE+FM+RM. For increases, the dark-blue segment reaches the AE+FM value and the light-blue segment extends the bar to the AE+FM+RM value. For decreases, the filled bar ends at the AE+FM+RM value, and an unfilled hatched segment extends to the original AE+FM value. Hatched segments indicate reductions and are not included in the final filled-bar height. Numeric labels give the absolute correlation values, not the sizes of the increments.

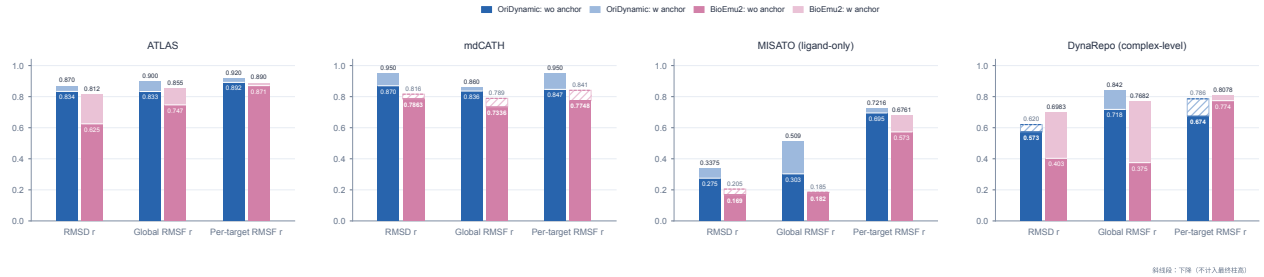


Figure 9. Effect of PTX-predicted versus supplied structural anchors. Panels are ordered as ATLAS, mdCATH, MISATO (ligand-only), and DynaRepo (complex-level). Each metric has two bars: blue for OriDynamic and pink for BioEmu2. Within each method, *wo anchor* uses a PTX-predicted structure, while *w anchor* uses the supplied reference structure. For increases, the saturated segment reaches the *wo anchor* value and the lighter segment extends to the *w anchor* value. For decreases, the filled bar ends at the *w anchor* value and the unfilled hatched segment marks the reduction from the *wo anchor* value; it is not included in the final filled-bar height. Labels report absolute correlation values. For OriDynamic, the *wo anchor* scores are identical to the AE+FM+RM scores in Fig. 8.

Global RMSF and Per-target RMSF. The effect is mixed for MISATO ligands: RMSD correlation decreases from 0.337 to 0.275, whereas Global RMSF and Per-target RMSF correlations increase from 0.291 to 0.303 and from 0.675 to 0.695, respectively. Overall, 11 of the 12 reported dataset–metric combinations increase after refinement, but the ligand RMSD result shows that this benefit is not uniform. These numerical differences alone do not establish statistical significance.

C.8.2 Effect of the structural-anchor source

We compare OriDynamic and BioEmu2 under two structural-reference settings. Here, *wo anchor* denotes inference without a supplied reference structure, using a PTX-predicted structure instead, whereas *w anchor* uses the supplied reference structure. Thus, *wo anchor* does not mean structure-free generation. In particular, OriDynamic retains its decoder anchor in this setting. Its *wo anchor* scores are the AE+FM+RM scores from the preceding comparison.

The effect of the reference source is method- and dataset-dependent (Fig. 9). For OriDynamic, all three correlations increase on ATLAS, mdCATH and MISATO when using a supplied reference. On DynaRepo, Global RMSF correlation increases from 0.718 to 0.842, but RMSD correlation decreases from 0.620 to 0.573 and Per-target RMSF correlation decreases from 0.786 to 0.674. For BioEmu2, all three correlations increase on ATLAS and DynaRepo but decrease on mdCATH. Its MISATO ligand Per-target RMSF correlation increases from 0.573 to 0.6761, while the other two correlations decrease. A supplied reference therefore does not uniformly increase the reported correlations across methods and datasets.